Beating Sampling: Mechanistic Estimation of the Failure Rate of a Neural Network at Frontier Scale

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Abstract

When can the failure rate of a neural network be estimated more efficiently by analyzing its internals than by sampling its outputs? We investigate this question for a class of random ReLU MLP "agents", and prove that the answer is "usually", with a single explicit estimation procedure which upper-bounds the failure rate p within a factor of 10, or else by at most 1%, for at least 99% of networks and random seeds. The result holds for every input dimension $n \geq 8{,}000$. For agents in our class with $8{,}000 \leq n \leq 16{,}384$ (comparable to the embedding width of current frontier transformers), the agent MLP has between $4.1 \cdot 10^{15}$ and $7.2 \cdot 10^{16}$ parameters, and the procedure costs about as much as a single exact sample of the agent's behavior. By contrast, consider a naive estimator that sees only the outcomes of exact samples and must succeed with probability 0.99 for every value of the failure rate p. When $p \leq 0.1\%$, the naive estimator needs more than 448 samples on average. For such low failure rates, our procedure is thus more than 447 times more efficient than naive sampling. The Beating Sampling Theorem and its proof are checked in Lean, and the translation of the Beating Sampling Theorem's assumptions and conclusions into Lean has been checked by GPT-6-Astra (high effort) and Claude Opus 5.5 (medium effort).

1 Introduction

This paper is the result of an experiment in human–LLM collaboration on mathematical interpretability research and exposition. While I, Andrew Critch, wrote the title and this introduction, the remainder of the content was written primarily by Claude-Opus-5.5 (medium effort, in Claude Code), with heavy editorial input from myself and GPT-6-Astra (high effort, in Codex). The paper describes technical work produced and formalized primarily by the same two models, in a collaboration enabled by javascript tools for them to send messages to each other. If you are primarily interested in how LLM agents were used to produce, formalize, and explain the result, please see Section 6.

I called our main result the Beating Sampling Theorem, because it shows how "white-box" statistical methods — or arguably, mechanistic interpretability techniques — are much more efficient than naive statistical sampling methods for estimating the frequency of rare events in neural network behavior. Notably, mechanistic analysis is applied not only to the "agent" networks, but to the world (or world model) from which their inputs are drawn.

I was inspired to pursue this question by the Alignment Research Center (ARC), and by my own nagging sense that the mechanistic interpretability of the world should also be represented in mathematical intepretability research, as well as the cost of sampling from or analyzing the world. Previous work by ARC in this area has represented and analyzed primarily the mechanisms and sampling costs of the agent, not the world. I also wanted to produce concrete, non-aysmptotic estimates of the computational costs and savings enabled by mechanistic analysis, at network scales somewhat comparable to current frontier AI models.

I'm not sure if everyone will consider the methods here "mechanistic" enough to be deserving of the term "mechanistic estimation". In case of objections, I would offer the term

"perspicacious estimation", as the estimation method peers into the innards of the network, and makes use of that insight to drastically reduce estimation costs.

2 Motivation

The statistical behavior of a neural network over an input distribution is usually estimated by sampling: collect inputs, evaluate the network on the inputs, count. However, behavior estimation by analyzing the internals of a network can be preferable to black-box sampling, for a variety of reasons:

1. **(sample count)** low-probability catastrophic events may require large numbers of samples to rule out, yielding high costs for eval-based safety assessments;
2. **(cost per sample)** for certain high-risk applications, access to real-world inputs is difficult, yielding a prohibitively high cost per sample; and
3. **(evaluation risk)** even in a testing environment, running a network with powerful cyberspace capabilities can compromise its containment, as in the infamous OpenAI–Huggingface intrusion.

By contrast, mechanistic interpretability methods aim to understand a network's behavior by analyzing its internal structure after training. Alignment problems may thus be addressed by looking for sources of malignment inside the network and ruling them out, or validating the presence of alignment mechanisms. If successful, mechanistic safety approaches could hence reduce all three of the costs above: large sampling counts, high costs per sample, and evaluation risks.

This note exhibits a process for estimating the failure rate of a random MLP, under a random failure metric and an arbitrary threshold. It follows on the work of the Alignment Research Center (ARC) on their "Matching Sampling Principle" (Neyman et al., 2025; Wu et al., 2026) as discussed in the Related Work section at the end. Unlike ARC's asymptotic width guarantees, our guarantees use explicit network dimensions, and we analyze a non-Gaussian input distribution generated by another, larger neural network rather than the Gaussian inputs of Wu et al. (2026).

3 Setting and result

MLP notation. For positive integers n, h, ℓ, n', let $\mathrm{MLP}(n, h, \ell, n')$ denote the set of multi-layer perceptrons (MLPs) with ReLU activation functions and no biases, with input dimension n, hidden width h, ℓ hidden layers and output dimension n'.

Network sizes. Rather than working asymptotically, our Beating Sampling Theorem will apply to agent MLPs above a certain size. Table 1 compares this size with the MLP components of open-weight frontier LLMs. Appendix A gives more details. Amongst these real-world models, $h \approx n^{1.6}$. The Beating Sampling Theorem uses $h = n^2$.

The agent and world networks. Consider an "agent" $A \in \mathrm{MLP}(n, n^2, 2, n)$. We also need some way of representing the complexity of the broader world from which the agent will receive inputs. For that, consider a much larger "world" network $W \in \mathrm{MLP}(2n, \lceil n^{3.5} \rceil, 2, n)$. This world and agent will not interact sequentially, so the "agent" is not very agent-like; these terms were chosen mainly to represent their relative sizes. Their dimensions were chosen somewhat arbitrarily to achieve about n^3 (roughly a trillion) times more parameters in the world network than in the agent network when we later prove the Beating Sampling Theorem. The world network will transform a high-entropy input u into a normalized input $W(u)/|W(u)|$ for the network A, where u is sampled uniformly from the unit sphere of

Network	n	h	ℓ	n'	Parameters
Kimi K3 MLPs (routed experts) (Moonshot AI, 2026)	3,584	2,752,512	1	3,584	$2.8 \cdot 10^{12}$
GLM-5.3 MLPs (Z.ai, 2026)	6,144	524,288	1	6,144	$7.53 \cdot 10^{11}$
DeepSeek V4 Pro MLPs (DeepSeek-AI, 2026)	7,168	1,179,648	1	7,168	$1.6 \cdot 10^{12}$
Qwen3.8-2.4T-A95B MLPs (Qwen Team, 2026)	8,192	1,048,576	1	8,192	$2.4 \cdot 10^{12}$
Muse Spark Open MLPs (Meta, 2026)	4,096	1,048,576	1	4,096	$8.16 \cdot 10^{11}$
Beating Sampling Theorem, Agent (A)	**8,000**	**64,000,000**	**2**	**8,000**	$\mathbf{4.1 \cdot 10^{15}}$
Beating Sampling Theorem, World (W)	**16,000**	$\mathbf{4.58 \cdot 10^{13}}$	**2**	**8,000**	$\mathbf{2.1 \cdot 10^{27}}$

Table 1: The agent and world networks of the Beating Sampling Theorem at $n = 8{,}000$, and the MLP blocks of large open-weight transformers. For a mixture-of-experts layer, h is the total width of its routed experts. The last column counts the parameters of the whole model; see Appendix A.

$\mathbb{R}^{2n}$. The normalization step reflects the practice of neural network developers choosing reasonably bounded units in which the networks receive their inputs from the world.

Now suppose the agent network A is being evaluated by a smaller "failure metric" network $F \in \mathrm{MLP}(n, \lceil n^{1.5} \rceil, 2, 1)$, with a threshold $t \in \mathbb{R}$ defining a Boolean failure classifier $C(y) \coloneqq \mathbf{1}[F(y) > t]$. Suppose the threshold t is arbitrary but fixed, and (W, A, F) are drawn independently with independent centered Gaussian weights, where the variance of each weight is $2/d$, and d is the number of inputs to the layer containing it (He initialization).

Given (W, A, F, t), we wish to estimate the failure probability

$$p \coloneqq \mathbb{P}_u \left[C\left(A\left(\frac{W(u)}{|W(u)|} \right) \right) = 1 \right]$$

with u sampled from the unit sphere as above. The Beating Sampling Theorem will give a concrete procedure $\mathrm{Est}(W, A, F, r; t)$, which will use the weights of W and A but never evaluate either network on an input; here r is a random seed. Even accounting for W, $\mathrm{Est}(W, A, F, r; t)$ will use fewer than $1.0003 \cdot 2n^7$ arithmetic operations, which is about the cost of one exact sample of $C(A(x))$. For every $n \geq 8{,}000$ and every t, for at least 99% of inputs (W, A, F, r), $\mathrm{Est}(W, A, F, r; t)$ will return an upper bound U with $p \leq U \leq \max(10p, 1/100)$. The success-probability and operation-count bounds of the Beating Sampling Theorem are proved in Lean, for the definitions and cost model used here.

Success and cost. Let $x_0 \coloneqq (1, 0, \ldots, 0) \in \mathbb{R}^n$ be the first standard basis vector. When $|W(u)| > 0$ let $x \coloneqq W(u)/|W(u)|$, else let $x \coloneqq x_0$. Then $p = \mathbb{P}_u[F(A(x)) > t]$. A program outputting U *succeeds* if $p \leq U \leq \max(10p, 1/100)$ [1]. Its *cost* is the number of operations it performs on real numbers: additions, subtractions, multiplications, divisions, square roots, ReLUs, comparisons, branches and table reads at computed indices; here $v/0 \coloneqq 0$, $\sqrt{v} \coloneqq$ the nonnegative square root of $\max(0, v)$, and $\mathrm{ReLU}(v) \coloneqq \max(0, v)$ for $v \in \mathbb{R}$. Reading an input (an entry of the seed r or a weight of W, A or F), a stored intermediate value or a rational constant costs nothing in the cost model in Lean [2], although future work could easily include these costs, as they are negligible (see the program below).

[1] `BstLean:Part157Defs,Part156Out`; Parts 29, 33, 35–36, 41, 48, 53, 56, 58, 73, 80, 105–108, 130–133, 135, 139–140, 142, 144–146, 156–157; CostModel.

[2] `BstLean:Part130Cost,Part282Join,Part288Search,Part289Run`.

Theorem (Beating Sampling Theorem). *Let $n \geq 8{,}000$ be an integer and $t \in \mathbb{R}$. Then $\mathrm{Est}(W, A, F, r; t)$ applies neither W nor A to any input, performs fewer than $1.0003 \cdot 2n^7$ operations, and for at least 99% of inputs (W, A, F, r) its output U satisfies*

$$p \;\leq\; U \;\leq\; \max(10p,\, 1/100).$$

For comparison, collecting a single $C(A(x))$ sample by evaluating W, A and F layer by layer needs at least $2n^7$ operations (Section 5). Both bounds of the Beating Sampling Theorem are checked in Lean, using only the axioms `propext`, `Classical.choice` and `Quot.sound` [1].

[1] `BstLean:Part293FinalAudit,Part289Cost`; declarations `Part293.classifier_stated2_8000` and `Part289.progOut_run2`; BSP.

4 Comparison with estimation by naive sampling

The Beating Sampling Theorem says that Est costs less than 1.0003 times as much as one exact sample of $C(A(x))$. How many samples would we need if we estimated p just by naively sampling and tallying C-values?

Drawing independent inputs u and naively evaluating $C(A(x))$ on each gives independent samples, like coin flips, each equal to 1 with probability p. Recall that a program outputting U *succeeds* if $p \le U \le \max(10p, 1/100)$ (Section 3). We call a program a *naive-99% sampler* if it sees only these flips and its own independent random numbers, decides after each flip whether to stop, and stops with probability 1 and succeeds with probability at least 0.99, for *every* value $p \in [0,1]$. Such a program is naive in three ways: it evaluates every network in full on every input; it ignores the weights of W, A and F, which it uses only through the flips; and it insists on succeeding with probability 0.99 for every p, instead of for 99% of inputs (W, A, F, r) as Est will.

Let N be the number of flips that a naive-99% sampler uses. For $s \in [0,1]$ and $v \in (0,1)$, let $\mathrm{KL}(s\|v) := s\ln(s/v) + (1-s)\ln((1-s)/(1-v))$, with $0\ln 0 := 0$. Say that an output U *succeeds at* a flip probability s if $s \le U \le \max(10s, 1/100)$.

Lemma 1. *Let $p \in [0,1]$ and $q \in (0,1)$, and suppose that no U succeeds both at p and at q. Then every naive-99% sampler satisfies $\mathbb{E}_p N \ge 0.98 \ln 99/\mathrm{KL}(p\|q)$.*

Proof. Lemma 1 extends to randomized samplers Wald's lower bound on the expected number of observations of a sequential test (Wald, 1945, Section 4.7, eq. (4.80)). In the present terms, Wald's bound states the following. Let a non-randomized sequential test between flip probabilities p and q stop with probability 1 at both, and let a and b be its error probabilities at p and at q. Then $\mathbb{E}_p N \cdot \mathrm{KL}(p\|q) \ge \mathrm{KL}(1-a\|b)$, where we extend KL to $v \in \{0,1\}$ by $s\ln(s/0) := \infty$ for $s > 0$ (Wald does not treat zero error probabilities; the bound holds trivially there). (Wald writes $-\mathrm{KL}(p\|q)$ as $E_0(z)$.)

The outputs that succeed at p and the outputs that succeed at q form two disjoint nonempty intervals, so one interval lies above the other. Let the sampler decide for the flip probability of the upper interval if U lies above the lower interval, and for the other flip probability otherwise. At each of p and q, success implies the correct decision, so both error probabilities a and b are at most 0.01. Let V be the sampler's own random numbers, which are independent of the flips. For almost every value v of V, the sampler with $V = v$ is a non-randomized sequential test that stops with probability 1 at both p and q. Let $a(v)$ and $b(v)$ be its error probabilities, so that $\mathbb{E}a(V) = a$ and $\mathbb{E}b(V) = b$. Wald's bound gives $\mathbb{E}_p[N \mid V = v] \cdot \mathrm{KL}(p\|q) \ge \mathrm{KL}(1-a(v)\|b(v))$. KL is jointly convex in its two arguments, so averaging over V gives $\mathbb{E}_p N \cdot \mathrm{KL}(p\|q) \ge \mathrm{KL}(1-a\|b)$. For $s \ge v$, $\mathrm{KL}(s\|v)$ increases in s and decreases in v; since $1-a \ge 0.99 \ge 0.01 \ge b$, this gives $\mathrm{KL}(1-a\|b) \ge \mathrm{KL}(0.99\|0.01) = 0.98\ln 99$. Finally, $p \ne q$, since otherwise $U = p$ would succeed at both, so $\mathrm{KL}(p\|q) > 0$. Hence $\mathbb{E}_p N \ge 0.98\ln 99/\mathrm{KL}(p\|q)$, as claimed. □

For $p \le 1/1000$, Lemma 1 implies that every naive-99% sampler needs more than 448 samples on average, as follows. In this range, $\max(10p, 1/100) = 1/100$, so success at p requires $U \le 1/100$. For every $q \in (1/100, 1)$, success at q requires $U \ge q > 1/100$. So Lemma 1 gives $\mathbb{E}_p N \ge 0.98\ln 99/\mathrm{KL}(p\|q)$, and letting q decrease to $1/100$ gives $\mathbb{E}_p N \ge 0.98\ln 99/\mathrm{KL}(p\|1/100)$. The denominator decreases in p on $[0, 1/100)$, so the bound is least at $p = 0$, where it equals $0.98\ln 99/\ln(100/99) > 448$. Hence, for $p \le 1/1000$, every naive-99% sampler needs more than 448 samples on average, as claimed. Each flip is a

naively computed exact sample of $C(A(x))$, which costs at least $2n^7$ operations [1]. So when $p \le 1/1000$, every naive-99% sampler has expected cost more than $448 \cdot 2n^7$, which is more than 447 times the cost bound $1.0003 \cdot 2n^7$ of Est.

[1] `BstLean:Part189Baseline`. The bounds from Wald's inequality and their numerical values were not checked in Lean.

Now, for $1/1000 < p < 1/10$, every naive-99% sampler needs more than 31 samples on average, and for $p \ge 1/10$ more than 1.95 samples, as follows. If $1/1000 < p < 1/10$, success at p requires $U \le 10p$, while success at q requires $U \ge q > 10p$ for every $q \in (10p, 1)$. If $p > 1/100$, success at p requires $U \ge p$, while success at q requires $U \le \max(10q, 1/100) < p$ for every $q \in (0, p/10)$. So Lemma 1 applies to these pairs. Letting q decrease to $10p$ or increase to $p/10$ in Lemma 1 gives $\mathbb{E}_p N \ge 0.98 \ln 99/\mathrm{KL}(p\|10p)$ or $\mathbb{E}_p N \ge 0.98 \ln 99/\mathrm{KL}(p\|p/10)$, respectively. Both bounds decrease in p. Hence $\mathbb{E}_p N > 31.16$ for $1/1000 < p < 1/10$ and $\mathbb{E}_p N > 1.955$ for $p \ge 1/10$, using the first bound up to $p = 1/100$ and the second above it, as claimed. Since each flip costs at least $2n^7$ operations, the expected cost of every naive-99% sampler is more than 31 times the cost bound of Est when $1/1000 < p < 1/10$, and more than 1.95 times it when $p \ge 1/10$.

In summary, every naive-99% sampler has a higher expected cost than Est at every p: more than 1.95 times for $p \ge 1/10$, more than 31 times for $1/1000 < p < 1/10$, and more than 447 times for $p \le 1/1000$. This trend underscores the importance of using statistical tests that are "perspicacious" in that they can read and use network weights, and/or "pragmatic" in that they work on a large fraction of inputs instead of all inputs, especially for low-probability high-cost events.

5 The program Est and the proof of the Beating Sampling Theorem

Est takes the input $(W, A, F, r; t)$ and follows the order of the networks' own computation, i.e., through W, then A, then F, with draws from the seed r in place of u [1]. Each step below says what Est computes, then analyzes it, stating each lemma where the analysis establishes it, and then says what the step costs [2]. The closing paragraphs combine the lemmas into the Beating Sampling Theorem.

[1] `BstLean:Part157Paper,Part169Pass,Part272Stated,Part289Design`; Parts 141, 150, 157, 159, 169, 201, 272, 278, 289.

[2] `BstLean:`
`Part282World,Part282Agent2,Part150Prog4,Part282RowData,Part282Grid,Part282NodeFull,`
`Part282NodeAny,Part282Draw,Part282Out,Part282JoinCost,Part289Hit,Part289Run,Part289Cost`.

Notation. For vectors, $|\cdot|$ is the Euclidean norm, $\langle \cdot, \cdot \rangle$ the inner product, and $\cos \angle(y, z) \coloneqq \langle y, z \rangle / (|y||z|)$. I is the identity matrix of the size the context requires. $\mathbb{S}^{k-1}$ is the unit sphere of $\mathbb{R}^k$. A subscript on $\mathbb{P}$ or $\mathbb{E}$ names the variables averaged over, with the others fixed. An unsubscripted $\mathbb{P}$ or $\mathbb{E}$ is over (W, A, F, r).

Weight matrices. With ReLU applied coordinatewise to real vectors, for $z \in \mathbb{R}^{2n}$ and $z' \in \mathbb{R}^n$ write

$$W(z) \coloneqq W_3 \,\mathrm{ReLU}(W_2 \,\mathrm{ReLU}(W_1 z)), \qquad A(z') \coloneqq A_3 \,\mathrm{ReLU}(A_2 \,\mathrm{ReLU}(A_1 z')),$$

and $m_W \coloneqq \lceil n^{3.5} \rceil$ [1].

[1] `BstLean:Part130Defs,Part157Triple`.

5.1 High-level overview

Lemma 2. *One exact sample of $C(A(x))$, computed by evaluating W, A and F layer by layer, costs at least $2n^7$ operations, which is more than $4.1 \cdot 10^{27}$ at $n = 8{,}000$.*

Proof. An exact sample computes the product of W_2 and the world's first hidden vector at u. Since W_2 is an $m_W \times m_W$ matrix, this product takes $2m_W^2 - m_W$ operations. The product $W_1 u$ takes more than m_W further operations. Hence an exact sample costs at least $2m_W^2 \geq 2n^7$ operations, since $m_W \geq n^{3.5}$, as claimed. □

Est multiplies W_2 by one vector only: an estimate of the world's mean first hidden vector. Est never applies W or A to an input.

Lemma 3. *Let M be a random matrix with independent $\mathcal{N}(0, \sigma^2)$ entries, and let D be random data independent of M. Let g be a fixed measurable function from the values of D to vectors, and let $v := g(D)$. Let Π_v be the orthogonal projection onto the directions orthogonal to v, i.e., $\Pi_v := I - vv^\top/|v|^2$ if $v \neq 0$, and $\Pi_0 := I$. Then, given D, the matrix $M\Pi_v$ is Gaussian and independent of Mv [1].*

Proof. Lemma 3 is standard; see Yang (2019, arXiv v3, Lemmas G.6–G.7), which extend Bayati and Montanari (2011, arXiv v4, Lemmas 11–12) to singular constraints. First let v be fixed. Each entry of $M\Pi_v$ and of Mv is a linear combination of the entries of M, so the pair $(M\Pi_v, Mv)$ is jointly Gaussian with mean zero. The rows of M are independent $\mathcal{N}(0, \sigma^2 I)$ vectors, so the cross-covariances are $\mathbb{E}[(M\Pi_v)_{ij}(Mv)_k] = \delta_{ik}\sigma^2(\Pi_v v)_j = 0$, since $\Pi_v v = 0$. Jointly Gaussian vectors with zero cross-covariance are independent, so $M\Pi_v$ is independent of Mv. Now let D be random. Since D is independent of M, the distribution of M given D is the distribution of M, and v is fixed given D. Hence, by the case of a fixed v applied to each value of D, given D the matrix $M\Pi_v$ is Gaussian and independent of Mv, as claimed. □

[1] `BstLean:Part134Split,Part157Reg`.

Lemma 3 applied first to the world and then to the agent implies that Est can avoid fully evaluating W or A on any inputs.

Applied to the world, Lemma 3 gives a proxy input, as follows. The only use Est makes of the last layer W_3 is to compute one vector $\hat{c} \in \mathbb{R}^n$. Step 2 will compute $\hat{c} := W_3 b_W$ for an estimate b_W of the world's mean last hidden vector. Call $\Omega_W := W_3\Pi_{b_W}$ the residual part of W_3. By Lemma 3, Ω_W is Gaussian, even given W_1, W_2 and $\hat{c}$, which are all that Est uses of W. So for each u, over Ω_W, the input x is the direction of the sum of a multiple of $\hat{c}$ and isotropic Gaussian noise. The scale of this noise depends on u. Est replaces this scale by an estimate τ_0 that is the same for every u. Hence, with this replacement, Lemma 3 applied to the world gives a *proxy input* $\tilde{x}$, whose distribution depends only on $\hat{c}$ and τ_0, which Est computes, as claimed. Lemma 7 in step 2 will state the distribution of x precisely.

Applied to the agent, Lemma 3 gives the distribution of the agent's output, as follows. The only use Est makes of A_3 is to compute $c_A := A_3 b_A$ for an estimate b_A of the agent's mean last hidden vector (step 3). Call $\Omega_A := A_3\Pi_{b_A}$ the residual part of A_3. By Lemma 3, Ω_A is Gaussian, even given W, A_1, A_2 and c_A. Hence, over Ω_A, the agent's output at each input is the sum of a multiple of c_A and isotropic Gaussian noise, as claimed. Lemma 9 in step 3 will state the distribution of the agent's output precisely.

Est computes b_W and b_A from the earlier layers alone, so Lemma 3 applies to b_W and b_A. The proof will not assume that b_W and b_A are accurate. Their accuracy will enter only through the error bounds Δ_W and Δ_A of the chain (1) below and through the frame bounds of Lemma 20.

At the proxy input, the coefficient of c_A and the variance of the noise in the agent's output of Lemma 9 depend mainly on one number: the component of $\tilde{x}$ along $\hat{c}/|\hat{c}|$, which step 4 will call the *height* of $\tilde{x}$. So steps 4–5 will tabulate the coefficient and the variance at $512n$ heights that depend only on the data $\mathcal{D}$ defined below, called *nodes*. Step 6 will then compute $K = 10^{12}$ *proxy outputs* from the tables and the seed r. Est evaluates F once

on each proxy output, instead of evaluating W, A and F once per exact sample. Finally, step 7 will scale and clip the fraction R of proxy outputs that fail, and output the result.

Let $\mathcal{D} \coloneqq (W_1, W_2, \hat{c}, A_1, A_2, c_A, F)$ be the data that Est uses apart from the seed r; Est reads W_3 and A_3 only to compute $\hat{c}$ and c_A. Let $\tilde{p}$, which step 3 will define, be the failure probability at the proxy input, and let $\tilde{q}$, which step 6 will define, be the probability that a proxy output fails. The proof follows a chain of approximations from p to R:

$$p \underset{\text{conc.}}{\approx} \mathbb{E}_{\Omega_W} p \underset{\Delta_W}{\approx} \tilde{p} \underset{\text{conc.}}{\approx} \mathbb{E}_{\Omega_A} \tilde{p} \underset{\Delta_A}{\approx} \tilde{q} \underset{\Delta_s}{\approx} R. \tag{1}$$

The two links marked "conc." will be concentration bounds over Ω_W and Ω_A. The other three will be bounds on the *square-root scale*, i.e., of the form $|\sqrt{a} - \sqrt{b}| \le \Delta$, with errors Δ_W, Δ_A and Δ_s. Step 6 will prove the Δ_A link, and step 7 the Δ_s link. The paragraph "Bounds over the residual parts" will prove the Δ_W link and the concentration bounds for the two links marked "conc.", on a good set $\mathcal{G}$ of values of $\mathcal{D}$. The paragraph "Assembly" will combine all five links.

5.2 Detailed overview

1. **Constants for the layers of W.** This step computes constants that depend only on n.

   Computation: Let $N \coloneqq 2n$. For any integer $T \ge 1$ and $v \in \mathbb{R}$, let P_T and Q_T be the truncated Taylor series of arctan and arcsin, i.e.,

   $$P_T(v) \coloneqq \sum_{0 \le j < T} \frac{(-1)^j v^{2j+1}}{2j+1}, \qquad Q_T(v) \coloneqq \sum_{0 \le j < T} \binom{2j}{j} \frac{v^{2j+1}}{4^j (2j+1)}.$$

   Let T_π be the least integer $T \ge 1$ with $25^T \ge 10^{23} N^2$. Compute Machin's approximation of π, i.e., $\tilde{\pi} \coloneqq 16 P_{T_\pi}(1/5) - 4 P_{T_\pi}(1/239)$. Compute an approximation of the mean of $\mathrm{ReLU}(u_1)$, i.e., $\tilde{c} \coloneqq \tilde{\pi}^{-1} \prod_{j=1}^{n-1} \frac{2j}{2j+1}$. Compute an approximation of twice the variance of $\mathrm{ReLU}(u_1)$, i.e., $\tilde{d} \coloneqq N^{-1} - 2\tilde{c}^2$. Compute an approximation of the correlation of the first-layer features of two independent inputs, i.e., $\tilde{r} \coloneqq 2N\tilde{c}^2$. Let T_κ be the least integer $T \ge 1$ with $0.355 \cdot 0.1014^T \le 10^{-5}/n^2$. Compute an approximation of the first-order arc-cosine kernel at $\tilde{r}$, i.e.,

   $$\tilde{\kappa} \coloneqq \big(\sqrt{1 - \tilde{r}^2} + \tilde{r}(\tilde{\pi}/2 + Q_{T_\kappa}(\tilde{r}))\big)/\tilde{\pi}.$$

   Analysis: The constant $\tilde{\pi}$ truncates the series of Machin's formula $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and the error $|\tilde{\pi} - \pi|$ falls geometrically in T_π.

   Lemma 4. $|\tilde{\pi} - \pi| \le 4 \cdot 25^{-T_\pi}$ *[1].*

   Proof. Lemma 4 is standard: its proof combines Machin's formula with a bound on the tail of the Taylor series of arctan. Let $T \coloneqq T_\pi$ and $0 \le v < 1$. Since $\arctan v = \sum_{j \ge 0} (-1)^j v^{2j+1}/(2j+1)$, the tail after T terms is at most the geometric sum $\sum_{j \ge T} v^{2j+1}$ in absolute value, i.e., $|\arctan v - P_T(v)| \le v^{2T+1}/(1-v^2)$. By Machin's formula, $\tilde{\pi} - \pi = 16(P_T(1/5) - \arctan \frac{1}{5}) - 4(P_T(1/239) - \arctan \frac{1}{239})$. Hence, by this tail bound at $v = 1/5$ and at $v = 1/239$, and since $239^{-2T} \le 25^{-T}$,

   $$|\tilde{\pi} - \pi| \le \frac{16/5}{1 - 1/25}\, 25^{-T} + \frac{4/239}{1 - 1/239^2}\, 25^{-T} = \Big(\frac{10}{3} + \frac{956}{57{,}120}\Big) 25^{-T} < 4 \cdot 25^{-T},$$

   as claimed. □

   [1] `BstLean:Part157Paper,Part159Pi`.

   The constants $\tilde{c}$ and $\tilde{d}$ approximate $\mathbb{E}_u \mathrm{ReLU}(u_1)$ and $2 \operatorname{Var}_u \mathrm{ReLU}(u_1)$, respectively.

Lemma 5. *Let* $c \coloneqq \pi^{-1}\prod_{j=1}^{n-1}\frac{2j}{2j+1}$ *and* $d \coloneqq N^{-1}-2c^2$*, i.e.,* $\tilde{c}$ *and* $\tilde{d}$ *with* π *in place of* $\tilde{\pi}$*. Then* $c=\mathbb{E}_u\,\mathrm{ReLU}(u_1)$ *and* $d=2\,\mathrm{Var}_u\,\mathrm{ReLU}(u_1)$ *[1].*

Proof. Lemma 5 is an elementary computation. Since u is uniform on $\mathbb{S}^{N-1}$, the coordinate u_1 has density proportional to $(1-s^2)^{(N-3)/2}$ on $(-1,1)$. Substituting $s=\sin\theta$, and using $N-2=2n-2$, gives

$$\mathbb{E}_u\,\mathrm{ReLU}(u_1)=\frac{\int_0^{\pi/2}\sin\theta\cos^{2n-2}\theta\,d\theta}{2\int_0^{\pi/2}\cos^{2n-2}\theta\,d\theta}=\frac{1/(2n-1)}{\pi\prod_{j=1}^{n-1}\frac{2j-1}{2j}},$$

by Wallis' formula $\int_0^{\pi/2}\cos^{2n-2}\theta\,d\theta=\frac{\pi}{2}\prod_{j=1}^{n-1}\frac{2j-1}{2j}$, which follows from DLMF (2026, eqs. 5.12.1–5.12.2), with $\Gamma(1/2)=\sqrt{\pi}$ and $\Gamma(v+1)=v\Gamma(v)$. Since $(2n-1)\prod_{j=1}^{n-1}\frac{2j-1}{2j+1}=1$, the right side of the display equals c, so $\mathbb{E}_u\,\mathrm{ReLU}(u_1)=c$. Next, $\mathrm{ReLU}(s)^2+\mathrm{ReLU}(-s)^2=s^2$ for $s\in\mathbb{R}$, the map $u\mapsto -u$ preserves the distribution of u, and $\mathbb{E}_u u_1^2=1/N$, since $\sum_i u_i^2=1$ and the coordinates of u have the same distribution. So $\mathbb{E}_u\,\mathrm{ReLU}(u_1)^2=1/(2N)$. Hence $2\,\mathrm{Var}_u\,\mathrm{ReLU}(u_1)=N^{-1}-2c^2=d$, as claimed. □

[1] `BstLean:Part159CN,Part137Hat2`. Lean checks the formula for c and the second moment $\mathbb{E}_u\,\mathrm{ReLU}(u_1)^2=1/(2N)$; the identity for d then follows by the last line of the proof.

The constant $\tilde{\kappa}$ approximates the first-order arc-cosine kernel of Cho and Saul (2009, eqs. (3) and (6)) at unit vectors with inner product v, i.e., $\kappa(v)\coloneqq(\sqrt{1-v^2}+v(\pi/2+\arcsin v))/\pi$ for $v\in[-1,1]$, at $\tilde{r}$. $\tilde{\kappa}$ replaces π and arcsin by $\tilde{\pi}$ and Q_{T_κ}, with an error that shrinks as T_κ grows.

Cost: none for $\tilde{c}$, $\tilde{d}$ and $\tilde{\pi}$ in the cost model in Lean, which charges nothing for rational constants; the square root, addition and division that form $\tilde{\kappa}$ are charged in step 2. For $8{,}000\le n\le 16{,}384$, $T_\pi=23$ and $T_\kappa\le 14$. Computing $\tilde{c}$, $\tilde{\pi}$ and $\tilde{\kappa}$ would then take at most $4n+250$ operations: about $4n$ for the product in $\tilde{c}$, about $6T_\pi\le 150$ for $\tilde{\pi}$, and about $6T_\kappa\le 90$ for $\tilde{\kappa}$. So $4n+250\le 65{,}786$. Since $n\ge 8{,}000$, also $4n+250<10^{-23}\cdot 2n^7$.

2. **The proxy input.** This step computes the vector $\hat{c}$ and the proxy noise scale τ_0, which together define the proxy input. This step is the only one that multiplies by W_2.

   Computation: Let $v_1,\dots,v_{m_W}\in\mathbb{R}^N$ be the rows of W_1. Compute the estimated mean of each first-layer unit, i.e., $\psi_i\coloneqq\tilde{c}|v_i|$ for $1\le i\le m_W$. Compute the mean squared row norm, i.e., $\Sigma_W\coloneqq m_W^{-1}\sum_i|v_i|^2$. Use Σ_W to estimate the spread of the second-layer pre-activations, i.e., $\varsigma_W\coloneqq\sqrt{\Sigma_W\tilde{d}}$. For any integer $J\ge 1$ and any function $\Phi:\mathbb{R}\to\mathbb{R}$, let $B_J[\Phi]$ be the mean of Φ at a sum of J independent uniform random signs divided by $\sqrt{J}$, i.e.,

$$B_J[\Phi]\coloneqq\sum_{j=0}^{J}2^{-J}\binom{J}{j}\Phi\big((2j-J)/\sqrt{J}\big).$$

   Let $J_W\coloneqq\lceil\sqrt{m_W}\rceil$, and let $\nu_W\coloneqq 2/m_W$ be the variance of an entry of W_2 and W_3. Compute the estimated mean b_W of the world's last hidden layer, the image $\hat{c}$ of b_W under W_3, the estimated cosine ℓ_0 between a last hidden vector of the world and b_W, and the proxy noise scale τ_0, i.e.,

$$b_{W,i}\coloneqq B_{J_W}[z\mapsto\mathrm{ReLU}((W_2\psi)_i+\varsigma_W z)]\text{ for }1\le i\le m_W,\qquad \hat{c}\coloneqq W_3 b_W,$$
$$\ell_0\coloneqq\sqrt{\frac{\Sigma_W\tilde{\kappa}^2}{N\nu_W|b_W|^2}},\qquad \tau_0\coloneqq\sqrt{\nu_W|b_W|^2\max\big(0,\,\tfrac{1}{\ell_0^2}-1\big)}.$$

   Analysis: Let $h_W:\mathbb{R}^N\to\mathbb{R}^{m_W}$ be the map to the world's last hidden layer, i.e., $h_W\coloneqq\mathrm{ReLU}\circ W_2\circ\mathrm{ReLU}\circ W_1$. The estimates are computed layer by layer. The first-layer unit i has mean $\mathbb{E}_u\,\mathrm{ReLU}(v_i\cdot u)=|v_i|\,\mathbb{E}_u\,\mathrm{ReLU}(u_1)$, and ψ_i estimates this mean. The vector b_W estimates $\mathbb{E}_u h_W(u)$ by treating each second-layer pre-activation as a Gaussian with mean

$(W_2\psi)_i$ and variance ς_W^2. The rule B_J approximates each such Gaussian expectation: $B_J[\Phi] \to \mathbb{E}_Z\Phi(Z)$ as $J \to \infty$, for $Z \sim \mathcal{N}(0,1)$ and continuous Φ of polynomial growth. The arc-cosine kernel motivates ℓ_0^2 as an estimate of $\cos^2 \angle(h_W(u), b_W)$ for most u. The proof will use about ψ, b_W and ℓ_0 only the bound Δ_W, the frame bound of Lemma 20, and the events that define the good set $\mathcal{G}$ of "Bounds over the residual parts".

Let Ω_W be the residual part of W_3, i.e., $\Omega_W \coloneqq W_3\Pi_{b_W}$, with Π_{b_W} as in Lemma 3. For each u, split $h_W(u)$ along b_W, i.e., $h_W(u) = \lambda_W(u)b_W + h_W^\perp(u)$ with $\lambda_W(u) \coloneqq \langle h_W(u), b_W\rangle/|b_W|^2$ and $h_W^\perp(u) \coloneqq h_W(u) - \lambda_W(u)b_W$.

Lemma 6. *For every u, $W(u) = \lambda_W(u)\hat{c} + \Omega_W h_W^\perp(u)$.*

Proof. Since $h_W^\perp(u)$ is orthogonal to b_W, $W_3h_W^\perp(u) = \Omega_W h_W^\perp(u)$. Hence, since $W_3b_W = \hat{c}$, $W(u) = W_3h_W(u) = \lambda_W(u)\hat{c} + \Omega_W h_W^\perp(u)$, as claimed. □

Lemma 6 gives the distribution of x. For each u with $\lambda_W(u) > 0$, let the *world noise scale* be

$$\tau_W(u) \coloneqq \frac{\sqrt{\nu_W}\,|h_W^\perp(u)|}{\lambda_W(u)}.$$

Lemma 7. *For each u with $\lambda_W(u) > 0$, x is the direction of a noisy copy of $\hat{c}$, i.e., given W_1, W_2 and $\hat{c}$, the distribution of x over Ω_W is that of $(\hat{c} + \tau_W(u)\xi)/|\hat{c} + \tau_W(u)\xi|$, with $\xi \sim \mathcal{N}(0, I)$ in $\mathbb{R}^n$, and with x_0 in place of this quotient when $\hat{c} + \tau_W(u)\xi = 0$* [1].

Proof. Fix u with $\lambda_W(u) > 0$. The vector b_W is a fixed function of (W_1, W_2), and W_3 is independent of (W_1, W_2). So, by Lemma 3 with $M = W_3$ and $D = (W_1, W_2)$, given W_1 and W_2, the matrix $\Omega_W = W_3\Pi_{b_W}$ is Gaussian and independent of $\hat{c} = W_3b_W$. The rows of Ω_W are independent $\mathcal{N}(0, \nu_W\Pi_{b_W})$ vectors, and $\Pi_{b_W}h_W^\perp(u) = h_W^\perp(u)$, where $h_W^\perp(u)$ is a fixed function of (W_1, W_2). So given W_1, W_2 and $\hat{c}$, the second term of Lemma 6, $\Omega_W h_W^\perp(u)$, has the distribution $\mathcal{N}(0, \nu_W|h_W^\perp(u)|^2 I)$, i.e., the distribution of $\sqrt{\nu_W}\,|h_W^\perp(u)|\,\xi$. Hence $W(u)$ has the distribution of $\lambda_W(u)(\hat{c} + \tau_W(u)\xi)$. Multiplication by $\lambda_W(u) > 0$ changes neither the direction of a vector nor whether the vector is 0. Hence x has the distribution of $(\hat{c}+\tau_W(u)\xi)/|\hat{c}+\tau_W(u)\xi|$, with x_0 in place of this quotient when $\hat{c} + \tau_W(u)\xi = 0$, as claimed. □

[1] `BstLean:Part157Reg,Part157Gp,Part157Repl,Part132Cos`. Lean takes $x_0 = (1, 0, \ldots, 0)$, and states the lemma as the law of the direction of $\lambda_W(u)\hat{c} + \sigma_W(u)\gamma$, with $\sigma_W(u) \coloneqq \sqrt{\nu_W}\,|h_W^\perp(u)|$ and $\gamma \sim \mathcal{N}(0, I)$, and also averaged over u; that form needs no condition on $\lambda_W(u)$, and dividing by $\lambda_W(u) > 0$ gives the form of the lemma.

The world noise scale $\tau_W(u)$ depends on u only through a cosine, i.e.,

$$\tau_W(u)^2 = \nu_W|b_W|^2\big(\cos^{-2}\angle(h_W(u), b_W) - 1\big).$$

Replacing the squared cosine by ℓ_0^2 gives τ_0^2, which Est computes. Replacing $\tau_W(u)$ by τ_0 in Lemma 7 gives the *proxy input*, i.e., $\tilde{x} \coloneqq (\hat{c} + \tau_0\xi)/|\hat{c} + \tau_0\xi|$ if $\hat{c} + \tau_0\xi \neq 0$ and $\tilde{x} \coloneqq x_0$ otherwise, with $\xi \sim \mathcal{N}(0, I)$ in $\mathbb{R}^n$. Unlike the distribution of x, the distribution of $\tilde{x}$ does not depend on the residual part Ω_W: it depends only on $\hat{c}$ and τ_0, which Est computes. So Est can estimate averages over $\tilde{x}$ (step 5) and draw exact samples of its height (step 6). The bound Δ_W will cover the error of replacing x by $\tilde{x}$, and also the rare inputs with $\lambda_W(u) \le 0$.

Cost: $2m_W^2 + 4m_W J_W + O(nm_W)$ operations. Almost all of these operations compute the product $W_2\psi$, which costs $2m_W^2 - m_W$. At $n = 8{,}000$, $J_W = 6{,}767{,}177$ and $4m_W J_W/(2m_W^2) < 3 \cdot 10^{-7}$. The cost model in Lean does not charge for the weights $2^{-J_W}\binom{J_W}{j}$ of B_{J_W}. These weights a_j satisfy $a_0 = 2^{-J_W}$ and $a_{j+1} = a_j(J_W - j)/(j+1)$. Computing all these weights by this recursion would take about $3J_W$ operations, i.e., about $2 \cdot 10^7$ at $n = 8{,}000$ and $7 \cdot 10^7$ at $n = 16{,}384$, less than 10^{-20} times $2m_W^2$.

3. **The agent's output distribution.** This step computes the vector b_A along which Est splits the agent's last hidden layer, and the image c_A of b_A under A_3, as step 2 did for the world.

   Computation: Let $a_1, \dots, a_{n^2} \in \mathbb{R}^n$ be the rows of A_1, and let $\nu_A \coloneqq 2/n^2$ be the variance of an entry of A_2 and A_3. Compute the root-mean-square norm of the unnormalized proxy input, i.e., $\rho \coloneqq \sqrt{|\hat{c}|^2 + n\tau_0^2}$. For $1 \le i \le n^2$, let ϕ_i be the activation of first-layer unit i at the unnormalized proxy input, i.e., $\phi_i(z) \coloneqq \mathrm{ReLU}(a_i \cdot \hat{c} + \tau_0 |a_i| z)$ for $z \in \mathbb{R}$. Estimate the first two moments of ϕ_i at a standard Gaussian by $\hat{h}_i \coloneqq B_n[\phi_i]$ and $\hat{h}_i^{(2)} \coloneqq B_n[\phi_i^2]$. Estimate the spread of the second-layer pre-activations, i.e., $\hat{\varsigma} \coloneqq \sqrt{\nu_A(\sum_i \hat{h}_i^{(2)} - |\hat{h}|^2)}$. Compute the estimated mean b_A of the agent's last hidden layer and the image c_A of b_A under A_3, i.e.,

$$b_{A,j} \coloneqq \rho^{-1} B_n[z \mapsto \mathrm{ReLU}((A_2 \hat{h})_j + \hat{\varsigma} z)] \text{ for } 1 \le j \le n^2, \qquad c_A \coloneqq A_3 b_A.$$

   Analysis: Let $h : \mathbb{R}^n \to \mathbb{R}^{n^2}$ be the map to the agent's last hidden layer, i.e., $h \coloneqq \mathrm{ReLU} \circ A_2 \circ \mathrm{ReLU} \circ A_1$. Since A has no biases, $h(\tilde{x}) = h(\hat{c} + \tau_0\xi)/|\hat{c} + \tau_0\xi|$ whenever $\hat{c} + \tau_0\xi \neq 0$, and $\rho^2 = \mathbb{E}_\xi |\hat{c} + \tau_0\xi|^2$. Each first-layer pre-activation $a_i \cdot (\hat{c} + \tau_0\xi)$ has the distribution of $a_i \cdot \hat{c} + \tau_0 |a_i| Z$ with $Z \sim \mathcal{N}(0,1)$, and the second layer is as in step 2. So b_A is designed to approximate $\mathbb{E}_{\tilde{x}} h(\tilde{x})$.

   As for the world in step 2, split the agent's last hidden layer along b_A. For $z \in \mathbb{R}^n$, let $\lambda(z)$ be the coefficient of $h(z)$ along b_A, $h^\perp(z)$ the rest of $h(z)$, and $S(z)$ the noise variance, i.e.,

$$\lambda(z) \coloneqq \frac{\langle h(z), b_A \rangle}{|b_A|^2}, \qquad h^\perp(z) \coloneqq h(z) - \lambda(z) b_A, \qquad S(z) \coloneqq \nu_A |h^\perp(z)|^2.$$

   Let Ω_A be the residual part of A_3, i.e., $\Omega_A \coloneqq A_3 \Pi_{b_A}$, with Π_{b_A} as in Lemma 3.

   Lemma 8. *For every $z \in \mathbb{R}^n$, $A(z) = \lambda(z) c_A + \Omega_A h^\perp(z)$.*

   Proof. Since $h^\perp(z)$ is orthogonal to b_A, $A_3 h^\perp(z) = \Omega_A h^\perp(z)$. Hence, since $A_3 b_A = c_A$, $A(z) = A_3 h(z) = \lambda(z) c_A + \Omega_A h^\perp(z)$, as claimed. □

   Lemma 9. *The agent's output is c_A times a coefficient plus isotropic noise, i.e., let z be a fixed function of W, A_1, A_2, c_A and ξ, with values in $\mathbb{R}^n$. Then, given W, A_1, A_2, c_A and ξ, the distribution of $A(z)$ over Ω_A is that of $\lambda(z) c_A + \sqrt{S(z)}\,\eta$, with $\eta \sim \mathcal{N}(0, I)$ in $\mathbb{R}^n$* [1].

   Proof. Lemma 9 is the Gaussian conditioning trick with the single constraint $c_A = A_3 b_A$; see Yang (2019, arXiv v3, Lemma G.7). The vector b_A is a fixed function of W, A_1 and A_2, and A_3 is independent of (W, A_1, A_2, ξ). So, by Lemma 3 with $M = A_3$ and $D = (W, A_1, A_2, \xi)$, given W, A_1, A_2 and ξ, the matrix Ω_A is Gaussian and independent of $c_A = A_3 b_A$, and the rows of Ω_A are independent $\mathcal{N}(0, \nu_A \Pi_{b_A})$ vectors. So given W, A_1, A_2, c_A and ξ, the vector z is fixed, Ω_A has the same distribution, and, since $\Pi_{b_A} h^\perp(z) = h^\perp(z)$, the term $\Omega_A h^\perp(z)$ has the distribution $\mathcal{N}(0, \nu_A |h^\perp(z)|^2 I) = \mathcal{N}(0, S(z) I)$. Hence, by Lemma 8, $A(z)$ has the distribution of $\lambda(z) c_A + \sqrt{S(z)}\,\eta$ given W, A_1, A_2, c_A and ξ, as claimed. □

   [1] `BstLean:Part157Reg,Part157Net,Part157Triple`. Lean proves a joint form (`triple_disint`): $\mathcal{D}$ and the three networks have the law obtained by rebuilding W_3 and A_3 from $\mathcal{D}$ and from Gaussian matrices G_W and G_A, independent of each other and of $\mathcal{D}$, whose entries have variances ν_W and ν_A, respectively; the residual parts are $G_W \Pi_{b_W}$ and $G_A \Pi_{b_A}$. With `mlp2_reb`, which proves Lemma 8 at every input, and `resid_gauss`, this gives Lemma 9 at each input.

   The splitting of A_3 in Lemma 8 is the Gaussian conditioning trick of Bolthausen (2014), also used by Bayati and Montanari (2011) and by Yang (2019) in Tensor Programs. Let $\tilde{p}$ be the failure probability at the proxy input, i.e., $\tilde{p} \coloneqq \mathbb{P}_\xi[F(A(\tilde{x})) > t]$, a function of $\mathcal{D}$ and Ω_A.

Cost: $2n^4 + 20n^3 + O(n^2)$ operations, including $2n^3$ for $c_A = A_3 b_A$. Almost all of these operations compute the product $A_2\hat{h}$. The cost model in Lean does not charge for the weights of B_n. Computing these weights as in step 2 would take about $3n$ more operations.

4. **A grid of heights.** By Lemma 9, the agent's output at z is determined in distribution by $\lambda(z)$ and $S(z)$. This step defines the *nodes*: a grid of $512n$ heights. Step 5 will tabulate estimates of two hidden-layer averages at the nodes, and step 6 will use these estimates to estimate λ and S.

   Computation: Let e_1 be the first standard basis vector of $\mathbb{R}^n$, and let $\hat{u}$ be the direction of $\hat{c}$, i.e., $\hat{u} \coloneqq \hat{c}/|\hat{c}|$ if $\hat{c} \neq 0$, else $\hat{u} \coloneqq -e_1$. Call $z \cdot \hat{u}$ the *height* of a unit vector $z \in \mathbb{R}^n$. Compute the central height $s_c \coloneqq |\hat{c}|/\rho$ and the *nodes*, i.e., the $512n$ heights around s_c given by $s_g \coloneqq s_c + (6\tau_0/\rho)\big((2g+1)/(512n) - 1\big)$ for integers $0 \le g < 512n$. At $n = 8{,}000$, there are $512n = 4{,}096{,}000$ nodes. Compute the component of each row a_i along $\hat{u}$, and the length of the component of a_i orthogonal to $\hat{u}$, i.e., $\alpha_i \coloneqq a_i \cdot \hat{u}$ and $\alpha_i^\perp \coloneqq \sqrt{|a_i|^2 - \alpha_i^2}$ for $1 \le i \le n^2$. Compute the energy of b_A, i.e., $e_A \coloneqq |b_A|^2/n$, which steps 5–6 will use.

   Analysis: The distribution of ξ is invariant under all rotations, so the distribution of $\tilde{x}$ is invariant under the rotations that fix $\hat{u}$.

   Lemma 10. *Given $\mathcal{D}$, assume $\hat{c} \neq 0$ and $\tau_0 > 0$, and let s be the height of the proxy input $\tilde{x}$. Then $\tilde{x} = s\hat{u} + \sqrt{1-s^2}\,\chi$, with χ uniform on the unit sphere orthogonal to $\hat{u}$ and independent of s. So, given s, the first-layer pre-activation $a_i \cdot \tilde{x}$ has the distribution of $s\alpha_i + \sqrt{1-s^2}\,\alpha_i^\perp V$ for each i, with V a coordinate of a uniform point of $\mathbb{S}^{n-2}$* [1].

   Proof. The lemma rests on the polar decomposition of a standard Gaussian vector g: $g/|g|$ is uniform on the unit sphere and independent of $|g|$ (Vershynin, 2018, Exercise 3.3.7). Given $\mathcal{D}$, the vectors $\hat{c}$ and $\hat{u} = \hat{c}/|\hat{c}|$ and the scale τ_0 are fixed, and ξ is independent of $\mathcal{D}$. Write $\xi = \xi_1\hat{u} + \xi^\perp$ with $\xi_1 \coloneqq \xi \cdot \hat{u}$. Then $\xi_1 \sim \mathcal{N}(0,1)$, $\xi^\perp$ is a standard Gaussian vector in the hyperplane orthogonal to $\hat{u}$, and ξ_1 and $\xi^\perp$ are independent. Let $\chi \coloneqq \xi^\perp/|\xi^\perp|$, which is defined almost surely since $n \ge 2$. By the polar decomposition applied to $\xi^\perp$, χ is uniform on the unit sphere orthogonal to $\hat{u}$ and independent of $(\xi_1, |\xi^\perp|)$. Since $\hat{c} + \tau_0\xi = (|\hat{c}| + \tau_0\xi_1)\hat{u} + \tau_0|\xi^\perp|\,\chi$, the height s is a function of $(\xi_1, |\xi^\perp|)$, and so χ is independent of s. The coefficient of χ in $\tilde{x}$ is nonnegative, and $|\tilde{x}| = 1$, so the coefficient is $\sqrt{1-s^2}$. Finally, $a_i \cdot \chi = a_i^\perp \cdot \chi$, with $a_i^\perp \coloneqq a_i - \alpha_i\hat{u}$ and $|a_i^\perp| = \alpha_i^\perp$. Since the distribution of χ is invariant under the rotations that fix $\hat{u}$, $a_i^\perp \cdot \chi$ has the distribution of $\alpha_i^\perp V$. Hence $a_i \cdot \tilde{x}$ has the distribution of $s\alpha_i + \sqrt{1-s^2}\,\alpha_i^\perp V$ given s, as claimed. □

   [1] `BstLean:Part161Lat,Part272Reduce`.

   The height of $\tilde{x}$ lies near s_c, with spread about τ_0/ρ, and the nodes span six spreads on each side of s_c. λ and S vary little among proxy inputs of equal height. Step 6 will bound the variation of λ and S.

   Cost: $4n^3 + O(n^2)$ operations. Almost all of these operations compute the products $a_i \cdot \hat{u}$ and $|a_i|^2$.

5. **Tables by height.** So that step 6 can estimate λ and S, this step tabulates two quantities of the agent's last hidden vector $h(z)$: the alignment of $h(z)$ with b_A and the energy of $h(z)$, i.e.,

$$f(z) \coloneqq \frac{\langle h(z), b_A \rangle}{\sqrt{n}\,|b_A|}, \qquad \omega(z) \coloneqq \frac{|h(z)|^2}{n} \qquad \text{for } z \in \mathbb{R}^n.$$

   This step estimates the averages of f and ω over proxy inputs whose height is a node.

   Computation: Let $J_z \coloneqq \lceil n^{1.5} \rceil - 1$, and let V be as in step 4. For each node g and $1 \le i \le n^2$, let $X_{g,i}$ be first-layer unit i at height s_g, i.e., $X_{g,i} \coloneqq \mathrm{ReLU}(s_g\alpha_i + \sqrt{1-s_g^2}\,\alpha_i^\perp V)$. Compute estimates $\bar{h}_{g,i}$ and $\bar{h}^{(2)}_{g,i}$ of $\mathbb{E}_V X_{g,i}$ and $\mathbb{E}_V X_{g,i}^2$. For even n, use the exact closed

forms of these expectations. For odd n, use a finite arithmetic rule whose error the Numerical certificates will bound [1]. Estimate the spread of the second-layer pre-activations, i.e., $\varsigma_g \coloneqq \sqrt{\nu_A(\sum_i \bar{h}_{g,i}^{(2)} - |\bar{h}_g|^2)}$. For $1 \le j \le n^2$, let $\phi_{g,j}(z) \coloneqq \mathrm{ReLU}((A_2\bar{h}_g)_j + \varsigma_g z)$, and estimate the first two moments of last-hidden-layer unit j by $\beta_{g,j} \coloneqq B_{J_z}[\phi_{g,j}]$ and $\beta_{g,j}^{(2)} \coloneqq B_{J_z}[\phi_{g,j}^2]$. Store the estimates $\tilde{f}_g$ and $\tilde{\omega}_g$ of the averages of f and ω at height s_g, i.e.,

$$\tilde{f}_g \coloneqq \frac{1}{\sqrt{n}\,|b_A|}\sum_j \beta_{g,j} b_{A,j}, \qquad \tilde{\omega}_g \coloneqq \frac{1}{n}\sum_j \beta_{g,j}^{(2)}.$$

[1] `BstLean:Part272Stated,Part272Even,Part278Err`.

Analysis: This step tabulates f and ω because the coefficient and the noise variance are functions of them.

Lemma 11. *The coefficient and the noise variance of Lemma 9 are functions of f, ω and e_A, i.e., $\lambda(z) = f(z)/\sqrt{e_A}$ and $S(z) = \gamma_A(\omega(z) - f(z)^2)$ for every $z \in \mathbb{R}^n$, where $\gamma_A \coloneqq n\nu_A = 2/n$ [1].*

Proof. Fix $z \in \mathbb{R}^n$, and write h, $h^\perp$, λ, S, f and ω for their values at z. First let $b_A \neq 0$. Then $\sqrt{e_A} = |b_A|/\sqrt{n}$, so $f/\sqrt{e_A} = \langle h, b_A\rangle/|b_A|^2 = \lambda$. Since $h = \lambda b_A + h^\perp$ with $h^\perp$ orthogonal to b_A, $|h^\perp|^2 = |h|^2 - \lambda^2|b_A|^2 = |h|^2 - \langle h, b_A\rangle^2/|b_A|^2 = n(\omega - f^2)$. So $S = \nu_A|h^\perp|^2 = n\nu_A(\omega - f^2) = \gamma_A(\omega - f^2)$. Now let $b_A = 0$. Then $f/\sqrt{e_A} = 0 = \lambda$, since $v/0 = 0$; so $h^\perp = h$, and $S = \nu_A|h|^2 = \gamma_A\omega = \gamma_A(\omega - f^2)$. In both cases, $\nu_A = 2/n^2$, so $\gamma_A = 2/n$. Hence the two identities hold for every z, as claimed. □

[1] `BstLean:Part169Var,Part169Red`. Lean proves the identity for λ in the form $f(z) = \sqrt{e_A}\,\lambda(z)$ as `fX_eq_lam`, and the identity for S as `sA_eq`, which assumes $b_A \neq 0$.

By Lemma 10, when n is even, $\bar{h}_{g,i}$ and $\bar{h}_{g,i}^{(2)}$ are the first two moments of first-layer unit i over proxy inputs of height s_g; when n is odd, $\bar{h}_{g,i}$ and $\bar{h}_{g,i}^{(2)}$ estimate these moments. The second layer is estimated as in step 3. So $\tilde{f}_g$ and $\tilde{\omega}_g$ are designed to approximate the averages of f and ω over proxy inputs of height s_g.

Cost: $1024n^5 + O(n^{4.5})$ operations, i.e., $512n$ nodes at $2n^4 + O(n^{3.5})$ each. Almost all of these operations compute the products $A_2\bar{h}_g$. At $n = 8{,}000$, the leading term $1024n^5$ is $8 \cdot 10^{-6}$ times $2n^7$. The cost model in Lean does not charge for the weights of B_{J_z} or for the coefficients of the arithmetic rules for $\bar{h}_{g,i}$ and $\bar{h}_{g,i}^{(2)}$, such as $(-1)^q\binom{P}{q}/(2q + j + 1)$ with $P \coloneqq (n-4)/2$ for even n. Computing these weights and coefficients once would take about $3J_z + O(n)$ operations, i.e., about $2 \cdot 10^6$ at $n = 8{,}000$, less than 10^{-16} times $1024n^5$.

6. **Proxy outputs.** This step draws the heights of K proxy inputs. For each proxy input, this step draws a proxy output from the distribution of Lemma 9, with λ and S replaced by their tabulated estimates.

   Computation: To keep the noise variance positive, Est uses a floor S_0 for $\tilde{\omega}_g - \tilde{f}_g^2$: let $S_0 \coloneqq 0.17$ if $n < 11{,}883$ and $S_0 \coloneqq 0.2$ otherwise. Read the seed as $r = (r_k)_{k=1}^K$ with $K \coloneqq 10^{12}$ and independent standard Gaussian vectors $r_k = (Z_k, G_{k,1}, \ldots, G_{k,n-1}, \eta_k) \in \mathbb{R} \times \mathbb{R}^{n-1} \times \mathbb{R}^n$. For each $k = 1, \ldots, K$, compute the height s_k of the k-th proxy input from the coordinate w_k of the unnormalized proxy input along $\hat{u}$, i.e.,

$$w_k \coloneqq |\hat{c}| + \tau_0 Z_k, \qquad s_k \coloneqq \frac{w_k}{\sqrt{w_k^2 + \tau_0^2\sum_i G_{k,i}^2}}.$$

   Let $g_k \in \{0, \ldots, 512n - 1\}$ index the node nearest to s_k, the larger at a tie. Compute the coefficient $\tilde{\lambda}_k$ and the noise variance $\tilde{S}_k$ from $\tilde{f}_{g_k}$ and $\tilde{\omega}_{g_k}$ by Lemma 11, and form

the proxy output y_k, i.e.,

$$\tilde{\lambda}_k \coloneqq \frac{\tilde{f}_{g_k}}{\sqrt{e_A}}, \qquad \tilde{S}_k \coloneqq \gamma_A \max(S_0, \tilde{\omega}_{g_k} - \tilde{f}_{g_k}^2), \qquad y_k \coloneqq \tilde{\lambda}_k c_A + \sqrt{\tilde{S}_k}\,\eta_k.$$

Analysis: The vector $(w_k, \tau_0 G_{k,1}, \ldots, \tau_0 G_{k,n-1})$ has the distribution of the unnormalized proxy input $\hat{c} + \tau_0 \xi$ in coordinates whose first axis is $\hat{u}$.

Lemma 12. *Given $\mathcal{D}$, if $\hat{c} \neq 0$ or $\tau_0 > 0$, then s_k has exactly the distribution of the height $\tilde{x} \cdot \hat{u}$ [1].*

Proof. The lemma rests on the rotation invariance of the standard Gaussian distribution (Vershynin, 2018, Proposition 3.3.2). Given $\mathcal{D}$, the vectors $\hat{c}$ and $\hat{u}$ and the scale τ_0 are fixed, and $\hat{c} = |\hat{c}|\hat{u}$, also when $\hat{c} = 0$. Let $\mathcal{Q}$ be an orthogonal $n \times n$ matrix with first column $\hat{u}$, and let $\xi' \coloneqq \mathcal{Q}^\top \xi$. Then $\xi' \sim \mathcal{N}(0, I)$, and $\mathcal{Q}^\top(\hat{c} + \tau_0 \xi) = (|\hat{c}| + \tau_0 \xi'_1, \tau_0 \xi'_2, \ldots, \tau_0 \xi'_n)$. Since $\mathcal{Q}^\top$ preserves lengths and maps $\hat{u}$ to e_1,

$$\tilde{x} \cdot \hat{u} = \frac{|\hat{c}| + \tau_0 \xi'_1}{\sqrt{(|\hat{c}| + \tau_0 \xi'_1)^2 + \tau_0^2 \sum_{i=2}^{n} \xi_i'^2}}.$$

The right side is the formula for s_k, with ξ' in place of $(Z_k, G_{k,1}, \ldots, G_{k,n-1})$. Both of these vectors have the distribution $\mathcal{N}(0, I)$. If $\tau_0 > 0$, the denominator is 0 only on an event of probability 0; if $\tau_0 = 0$, then $\hat{c} \neq 0$ and both $\tilde{x} \cdot \hat{u}$ and s_k equal 1. Hence s_k has the distribution of $\tilde{x} \cdot \hat{u}$, as claimed. □

[1] `BstLean:Part161Lat`. Lean proves the lemma as part of `nuStand_latitude`, which assumes $\hat{c} \neq 0$ and $\tau_0 > 0$.

Flooring $\tilde{\omega}_{g_k} - \tilde{f}_{g_k}^2$ at S_0 does not increase the absolute error of this difference as an estimate of a target value that is at least S_0.

Lemma 13. *Flooring never increases the distance to a target value $Y \geq S_0$, i.e., $|\max(S_0, v) - Y| \leq |v - Y|$ for reals v and $Y \geq S_0$ [1].*

Proof. If $v \geq S_0$, then $\max(S_0, v) = v$, and $|\max(S_0, v) - Y| = |v - Y|$. If $v < S_0$, then $\max(S_0, v) = S_0$ and $v < S_0 \leq Y$, so $|\max(S_0, v) - Y| = Y - S_0 < Y - v = |v - Y|$. Hence $|\max(S_0, v) - Y| \leq |v - Y|$ for every real v, as claimed. □

Let $\tilde{q}$ be the probability that one proxy output fails, i.e., $\tilde{q}(\mathcal{D}) \coloneqq \mathbb{P}_r[F(y_1) > t \mid \mathcal{D}]$. For probability densities p_1 and p_2, let the *Hellinger distance* be $H(p_1, p_2) \coloneqq \left(\int (\sqrt{p_1} - \sqrt{p_2})^2\right)^{1/2}$, with no factor $1/2$ inside the root. Two Gaussians $\mathcal{N}(\lambda c_A, SI)$ with nearby (λ, S) are close in Hellinger distance. Among proxy inputs of equal height, λ and $\sqrt{S}$ deviate little from their conditional means; Lean bounds moments of these deviations, averaged over the height, and the mixture comparison also accounts for exceptional directions [2]. A Hellinger bound on the distribution of the output bounds the difference of the square roots of the failure probabilities. This gives the following lemma.

Lemma 14. *For every $t \in \mathbb{R}$ and on the good set $\mathcal{G}$ of "Bounds over the residual parts", the average of $\tilde{p}$ over Ω_A is close to $\tilde{q}(\mathcal{D})$ on the square-root scale, i.e.,*

$$\left|\sqrt{\mathbb{E}_{\Omega_A} \tilde{p}} - \sqrt{\tilde{q}(\mathcal{D})}\right| \leq \Delta_A \text{ [2]},$$

where $\Delta_A > 0$ is an explicit real depending only on n, given in the Numerical certificates.

Proof sketch. The full proof is in Lean. Fix t and $\mathcal{D} \in \mathcal{G}$, and let $B \coloneqq \{y \in \mathbb{R}^n : F(y) > t\}$. The proof combines three standard inequalities. The first inequality is that

$$\left(\sqrt{P_1(B)} - \sqrt{P_2(B)}\right)^2 \leq H(p_1, p_2)^2 \tag{2}$$

for distributions P_1 and P_2 with densities p_1 and p_2, by the Cauchy–Schwarz inequality on B and on $\mathbb{R}^n \setminus B$. The second inequality is a bound on the Hellinger distance between two isotropic Gaussians. By the closed form of Devroye et al. (2022, Section 2), multiplied by 2 because H has no factor $1/2$ inside the root, with $\varrho \coloneqq S_1/S_2$,

$$\begin{aligned} H\big(\mathcal{N}(m_1, S_1 I), \mathcal{N}(m_2, S_2 I)\big)^2 &= 2 - 2\Big(\frac{2\sqrt{S_1 S_2}}{S_1 + S_2}\Big)^{n/2} e^{-|m_1 - m_2|^2/(4(S_1+S_2))} \\ &\le \frac{n(\varrho - 1)^2}{8\varrho} + \frac{|m_1 - m_2|^2}{2(S_1 + S_2)} \end{aligned} \tag{3}$$

for $S_1, S_2 > 0$, where the inequality follows from $1 - y \le -\ln y$ and $\ln y \le y - 1$ for $y > 0$. The third inequality is that

$$\big(\sqrt{\mathbb{E}X} - \sqrt{\mathbb{E}Y}\big)^2 \le \mathbb{E}\big(\sqrt{X} - \sqrt{Y}\big)^2 \tag{4}$$

for random variables X, Y with values in $[0, 1]$, by the Cauchy–Schwarz inequality.

By Lemma 9, $\mathbb{E}_{\Omega_A}\tilde{p}$ is the average over $\tilde{x}$ of $\mathcal{N}(\lambda(\tilde{x})c_A, S(\tilde{x})I)(B)$. By Lemma 12, $\tilde{q}(\mathcal{D})$ is the average over the height s of $\tilde{x}$ of the same Gaussian probability, with the tabulated coefficient and noise variance at the node nearest s in place of $\lambda(\tilde{x})$ and $S(\tilde{x})$. The Lean proof bounds the difference of the square roots of these two averages by a sum of four terms. The first term is the fluctuation of (λ, S) among proxy inputs of equal height, bounded through the χ^2 divergence of a Gaussian mixture. The second term is the bias of the tables against the averages of (λ, S) at each height, bounded by (2)–(4) and by Lemma 13. The third term comes from the heights outside the range of the nodes. The fourth term is the error of the rule of step 5 for odd n, at most 10^{-20}. The Numerical certificates will bound the sum of the four terms by Δ_A, as claimed. □

[1] `BstLean:Part289Floor`.
[2] `BstLean:Part157Repl,Part161Root,Part161Gauss,Part161Red,Part161Fl,Part281Odd, Part289Root,Part293Root,Part293Arithmetic`; Parts 161–171, 202, 207, 212–213, 216, 220, 223, 228, 239, 241–242, 248, 289; Parts 290–293.

Cost: at most $K(5n+10\lceil\log_2(512n)\rceil+27)$ operations, i.e., about $4{\cdot}10^{16}$ at $n = 8{,}000$. The cost model in Lean charges nothing for the seed. Generating the seed's $2nK$ Gaussians from random bits would take about $200Kn$ operations at an estimated 100 operations per Gaussian (e.g., by the Box–Muller method with series for ln, cos and sin), i.e., about $1.6 \cdot 10^{18}$ at $n = 8{,}000$. This generation cost is about 40 times the charged cost of this step, but less than $4 \cdot 10^{-10}$ times $2n^7$, and less than $2 \cdot 10^{-6}$ times the cost of step 7.

7. **The failure fraction and the output.** This step evaluates F on the proxy outputs. This step then scales and clips the fraction of proxy outputs that fail, which gives the output U.

   Computation: Let R be the fraction of proxy outputs that fail, i.e.,

$$R \coloneqq K^{-1} \sum_{k=1}^{K} \mathbf{1}[F(y_k) > t].$$

   Let $\Gamma \coloneqq 2.2524763228$. Output the scaled and clipped fraction, i.e.,

$$U \coloneqq \min(1, \max(1/100, \Gamma R)).$$

   Analysis: Est succeeds whenever R lies in the *window* $[p/\Gamma,\ \max(10p, 1/100)/\Gamma]$. The lower end matters only when $p > 1/100$, since otherwise $U \ge 1/100 \ge p$ by the definition of U. Since $1/\sqrt{\Gamma} < 0.6664$ and $\sqrt{10/\Gamma} > 2.1070$, the window gives the following lemma on the square-root scale.

Lemma 15. *If $p < 1/1000$,* Est *succeeds whenever $R \le 1/(100\Gamma)$. If $p \ge 1/1000$,* Est *succeeds whenever $0.667\sqrt{p} \le \sqrt{R} \le 2.107\sqrt{p}$ [1].*

Proof. Recall that $0 \le p \le 1$ and $R \ge 0$. First let $p < 1/1000$ and $R \le 1/(100\Gamma)$. Then $\Gamma R \le 1/100$, so $U = 1/100$, and $p < U \le \max(10p, 1/100)$. Now let $p \ge 1/1000$ and $0.667\sqrt{p} \le \sqrt{R} \le 2.107\sqrt{p}$. Since $0.667^2\Gamma > 1.0021$ and $2.107^2\Gamma < 9.9998$, squaring and multiplying by Γ give $p \le \Gamma R \le 10p$. So $U \ge \min(1, \Gamma R) \ge \min(1, p) = p$, and $U \le \max(1/100, \Gamma R) \le \max(1/100, 10p)$. Hence Est succeeds in both cases, as claimed. □

[1] `BstLean:Part156Out`. Lean proves, as `not_success`, the contrapositive of a slightly stronger form of the statement that Est succeeds whenever R lies in the window, for a factor $a > 0$ in place of Γ, and with the window stated for a value Q near R on the square-root scale, i.e., $|\sqrt{R} - \sqrt{Q}|$ at most a given error. The lemma follows with $a = \Gamma$, $Q = R$ and error 0, by the inequalities for $0.667^2\Gamma$ and $2.107^2\Gamma$ in its proof.

Given $\mathcal{D}$, the K proxy outputs fail independently, each with probability $\tilde{q}(\mathcal{D})$. A Chernoff bound then keeps $\sqrt{R}$ near $\sqrt{\tilde{q}}$.

Lemma 16. *If, given $\mathcal{D}$, $KR \sim \mathrm{Bin}(K, \tilde{q}(\mathcal{D}))$, then, for every real $\Delta_s > 0$, $\sqrt{R}$ is close to $\sqrt{\tilde{q}(\mathcal{D})}$, i.e., $\mathbb{P}_r(|\sqrt{R} - \sqrt{\tilde{q}(\mathcal{D})}| > \Delta_s \mid \mathcal{D}) \le 2e^{-K\Delta_s^2}$ [1].*

Proof. Lemma 16 is standard; see Okamoto (1958, Theorems 3–4), which give $e^{-2K\Delta_s^2} + e^{-K\Delta_s^2}$ in place of $2e^{-K\Delta_s^2}$. Condition on $\mathcal{D}$, and let $q \coloneqq \tilde{q}(\mathcal{D})$. If $q \in \{0, 1\}$, then $R = q$ almost surely, and the probability is 0. Let $0 < q < 1$, and let $D(z\|q) \coloneqq z\ln(z/q) + (1-z)\ln((1-z)/(1-q))$ for $0 < z < 1$. By the Chernoff bound (Okamoto, 1958, Lemma 1), $\mathbb{P}_r(R \ge z \mid \mathcal{D}) \le e^{-KD(z\|q)}$ for $z \ge q$, and $\mathbb{P}_r(R \le z \mid \mathcal{D}) \le e^{-KD(z\|q)}$ for $z \le q$. By $\ln y \le y - 1$ at $y = \sqrt{q/z}$ and at $y = \sqrt{(1-q)/(1-z)}$,

$$-D(z\|q) \le 2\big(\sqrt{zq} - z\big) + 2\big(\sqrt{(1-z)(1-q)} - (1-z)\big) = -(\sqrt{z} - \sqrt{q})^2 - (\sqrt{1-z} - \sqrt{1-q})^2,$$

so $D(z\|q) \ge (\sqrt{z} - \sqrt{q})^2$. The event $\sqrt{R} > \sqrt{q} + \Delta_s$ is empty if $(\sqrt{q} + \Delta_s)^2 \ge 1$, and otherwise lies in $\{R \ge z_+\}$ with $z_+ \coloneqq (\sqrt{q} + \Delta_s)^2 \in (q, 1)$. The event $\sqrt{R} < \sqrt{q} - \Delta_s$ is empty if $\sqrt{q} \le \Delta_s$, and otherwise lies in $\{R \le z_-\}$ with $z_- \coloneqq (\sqrt{q} - \Delta_s)^2 \in (0, q)$. For $z = z_+$ and for $z = z_-$, $(\sqrt{z} - \sqrt{q})^2 = \Delta_s^2$, so each event has probability at most $e^{-K\Delta_s^2}$. Hence the union of the two events has probability at most $2e^{-K\Delta_s^2}$, as claimed. □

[1] `BstLean:Part156Samp,Part232World`.

With $K = 10^{12}$, the bound of Lemma 16 is $2e^{-100}$ at $\Delta_s = 10^{-5}$, the value Lean uses for $n \ge 11{,}883$, and $2e^{-25} < 10^{-9}$ at $\Delta_s = 5 \cdot 10^{-6}$, the value Lean uses for $8{,}000 \le n \le 11{,}882$. Errors on the square-root scale suit the window, as follows. For $0 \le \Delta \le \sqrt{p}$, if $|\sqrt{R} - \sqrt{p}| \le \Delta$, then $(1 - \Delta/\sqrt{p})^2 \le R/p \le (1 + \Delta/\sqrt{p})^2$. Also, $\sqrt{p} \ge 0.0316$ wherever $\max(10p, 1/100) = 10p$, i.e., $p \ge 1/1000$. Hence, if $p \ge 1/1000$ and $|\sqrt{R} - \sqrt{p}| \le 0.01$, then $0.467 \le R/p \le 1.733$, inside $[1/\Gamma, 10/\Gamma]$, the window's range for R/p, as claimed.

Cost: $K(2m_F^2 + 2nm_F + O(m_F))$ operations, i.e., one evaluation of F on each of the K proxy outputs, where $m_F \coloneqq \lceil n^{1.5} \rceil$ is the hidden width of F.

5.3 Completion of the proof

Bounds over the residual parts. This paragraph proves three bounds: the Δ_W link of the chain (1), and the concentration bounds for the two links marked "conc.". Given $\mathcal{D}$, the matrices Ω_W and Ω_A and the seed r are independent and Gaussian, so p is a function of $(\mathcal{D}, \Omega_W, \Omega_A)$ and $\tilde{p}$ is a function of $(\mathcal{D}, \Omega_A)$. All three bounds hold on a good set $\mathcal{G}$ of values of $\mathcal{D}$. To define $\mathcal{G}$, let $\delta_{\mathrm{hid}}, \mu_W, \delta_{\mathrm{ext}}, \Delta_W > 0$ be explicit reals that depend only on n; for an example, the Numerical certificates will give Δ_W and a bound on $\delta_{\mathrm{hid}} + \mu_W/\Delta_W^2 + \delta_{\mathrm{ext}}$. For each u, let H_u be the Hellinger distance of step 6 between the noisy copy of $\hat{c}$ in Lemma 7 and the noisy copy of $\hat{c}$ that defines the proxy input, i.e., $H_u \coloneqq H\big(\mathcal{N}(\hat{c}, \tau_W(u)^2 I), \mathcal{N}(\hat{c}, \tau_0^2 I)\big)$ if $\lambda_W(u) > 0$, $\tau_W(u) > 0$ and $\tau_0 > 0$, and $H_u \coloneqq \sqrt{2}$ otherwise. Let $\bar{\Delta}$ be the root mean

square of H_u, i.e., $\bar{\Delta}(\mathcal{D}) \coloneqq (\mathbb{E}_u H_u^2)^{1/2}$, a function of $\mathcal{D}$. Let $\mathcal{H}$ be a set of values of $\mathcal{D}$ and $\mathcal{E}$ a further set of values of $\mathcal{D}$, such that

$$\mathbb{P}(\mathcal{D} \notin \mathcal{H}) \le \delta_{\text{hid}}, \qquad \mathbb{E}[\mathbf{1}_{\mathcal{H}} \bar{\Delta}^2] \le \mu_W, \qquad \mathbb{P}(\mathcal{D} \in \mathcal{H} \setminus \mathcal{E}) \le \delta_{\text{ext}}.$$

On $\mathcal{H}$, the world's hidden layers satisfy explicit bounds, which only the Lean proof states. Let the good set be $\mathcal{G} \coloneqq \mathcal{H} \cap \mathcal{E} \cap \{\bar{\Delta} \le \Delta_W\}$.

The concentration bounds depend on the directions of the vectors $h^\perp(\tilde{x})$. Let ζ be the direction of $h^\perp$, i.e., $\zeta \coloneqq h^\perp/|h^\perp|$ if $h^\perp \neq 0$ and $\zeta \coloneqq 0$ otherwise, and let the frame constant of ζ be $L_A \coloneqq \|\mathbb{E}_{\tilde{x}} \zeta(\tilde{x}) \zeta(\tilde{x})^\top\|_{\text{op}}$. Let L_W be the same for the world, i.e., with $h_W^\perp$ and u in place of $h^\perp$ and $\tilde{x}$. For real $L > 0$ and $\sigma \in \mathbb{R}$, let $\chi_L(\sigma) \coloneqq (e^{\sigma L} - 1)/L$. Lemma 19 below will bound the moment generating functions of p and $\tilde{p}$, by Lemma 18, a Gaussian form of the geometric Brascamp–Lieb inequality (Bennett et al., 2008, Proposition 2.8). Lemma 20 will bound L_A, by the Gaussian Poincaré inequality and a Schur test on pairs of proxy inputs.

Lemma 17. *On $\mathcal{G}$, $\tilde{p}$ is close to the average of p over Ω_W, i.e.,*

$$\left|\sqrt{\mathbb{E}_{\Omega_W} p} - \sqrt{\tilde{p}}\right| \le \Delta_W \text{ for every } \Omega_A, \quad \text{and} \quad \mathbb{P}(\mathcal{D} \notin \mathcal{G}) \le \delta_{\text{hid}} + \mu_W/\Delta_W^2 + \delta_{\text{ext}}. \text{ [1]}$$

Proof. The first claim needs only $\bar{\Delta} \le \Delta_W$. Fix t, Ω_A, and $\mathcal{D}$ with $\bar{\Delta}(\mathcal{D}) \le \Delta_W$, and let $T \coloneqq \{z \in \mathbb{S}^{n-1} : F(A(z)) > t\}$. For each u, let P_u be the probability over Ω_W that $x \in T$. By Fubini's theorem, $\mathbb{E}_{\Omega_W} p = \mathbb{E}_u P_u$. Fix u, and let f and g be the densities of $\mathcal{N}(\hat{c}, \tau_W(u)^2 I)$ and $\mathcal{N}(\hat{c}, \tau_0^2 I)$, respectively. If $H_u = \sqrt{2}$, then $(\sqrt{P_u} - \sqrt{\tilde{p}})^2 \le 1 \le H_u^2$, since $P_u, \tilde{p} \in [0,1]$. Otherwise, by Lemma 7 and the definition of $\tilde{x}$, $P_u = \int_S f$ and $\tilde{p} = \int_S g$ for the set S of vectors whose direction lies in T, up to a null set. By the Cauchy–Schwarz inequality, $\int_S \sqrt{fg} \le \sqrt{P_u \tilde{p}}$, so $(\sqrt{P_u} - \sqrt{\tilde{p}})^2 \le \int_S (\sqrt{f} - \sqrt{g})^2 \le H_u^2$. By the Cauchy–Schwarz inequality again, $\left(\sqrt{\mathbb{E}_u P_u} - \sqrt{\tilde{p}}\right)^2 \le \mathbb{E}_u \left(\sqrt{P_u} - \sqrt{\tilde{p}}\right)^2 \le \mathbb{E}_u H_u^2 = \bar{\Delta}^2$. Hence $|\sqrt{\mathbb{E}_{\Omega_W} p} - \sqrt{\tilde{p}}| \le \bar{\Delta} \le \Delta_W$, and in particular on $\mathcal{G}$. For the second claim, $\mathcal{D} \notin \mathcal{G}$ only if $\mathcal{D} \notin \mathcal{H}$, or $\mathcal{D} \in \mathcal{H} \setminus \mathcal{E}$, or $\mathcal{D} \in \mathcal{H}$ and $\bar{\Delta} > \Delta_W$. By Markov's inequality, the last event has probability at most $\mathbb{E}[\mathbf{1}_{\mathcal{H}} \bar{\Delta}^2]/\Delta_W^2 \le \mu_W/\Delta_W^2$. Hence, by a union bound, $\mathbb{P}(\mathcal{D} \notin \mathcal{G}) \le \delta_{\text{hid}} + \mu_W/\Delta_W^2 + \delta_{\text{ext}}$, as claimed. □

Lemma 18 (Gaussian geometric Brascamp–Lieb inequality)**.** *Let μ be a Borel probability measure on a Euclidean space $\mathcal{Z}$, let $\zeta : \mathcal{Z} \to \mathbb{R}^m$ be measurable with $|\zeta(z)| \in \{0, 1\}$ for every z, and let $L > 0$ with $\int \zeta \zeta^\top d\mu \preceq L\, I$. Let G be an $n \times m$ matrix of independent $\mathcal{N}(0,1)$ entries, and let $Z \sim \mathcal{N}(0, I)$ in $\mathbb{R}^n$. Let $0 < a \le b$, and let $(f_z)_{z \in \mathcal{Z}}$ be a family of functions $f_z : \mathbb{R}^n \to [a, b]$, jointly measurable in z and in the argument of f_z, such that f_z is constant for every z with $\zeta(z) = 0$. Then*

$$\mathbb{E}_G \exp\Big(L^{-1} \int \ln f_z(G\zeta(z))\, d\mu(z)\Big) \le \exp\Big(L^{-1} \int \ln \mathbb{E}_Z f_z(Z)\, d\mu(z)\Big) \text{ [2].}$$

Proof. Lemma 18 follows from the Boué–Dupuis formula (Lehec, 2014, Proposition 1.1), by the argument with which Lehec (2014, Section 2) proves the Brascamp–Lieb inequality. □

Lemma 19. *Given $\mathcal{D}$, let $L'_W \ge L_W$ and $L'_A \ge L_A$ be positive reals. Then p and $\tilde{p}$ concentrate around their averages over Ω_W and Ω_A, respectively, i.e.,*

$$\mathbb{E}_{\Omega_W} e^{\sigma p} \le e^{\chi_{L'_W}(\sigma)\, \mathbb{E}_{\Omega_W} p}, \qquad \mathbb{E}_{\Omega_A} e^{\sigma \tilde{p}} \le e^{\chi_{L'_A}(\sigma)\, \mathbb{E}_{\Omega_A} \tilde{p}}$$

for every $\sigma \in \mathbb{R}$ and, in the first bound, every Ω_A [2].

Proof. Consider $\tilde{p}$. The bound for p is the same, for each fixed Ω_A, with u, $h_W^\perp$, ν_W, L'_W and W_3 in place of $\tilde{x}$, $h^\perp$, ν_A, L'_A and A_3. Let $L \coloneqq L'_A$, and let μ be the distribution of $\tilde{x}$, so that $\int \zeta \zeta^\top d\mu \preceq L\, I$. As in the proof of Lemma 9, given $\mathcal{D}$, the matrix Ω_A has the distribution of $\sqrt{\nu_A}\, G \Pi_{b_A}$, with G an $n \times n^2$ matrix of independent $\mathcal{N}(0,1)$ entries. So $\Omega_A h^\perp(z) = \sqrt{\nu_A}\, |h^\perp(z)|\, G\zeta(z)$ for every z, jointly in distribution, and $\tilde{p} = \int \phi_z(G\zeta(z))\, d\mu(z)$, with $\phi_z(y) \coloneqq \mathbf{1}[F(\lambda(z) c_A + \sqrt{\nu_A}\, |h^\perp(z)|\, y) > t]$. The function ϕ_z is constant where $\zeta(z) = 0$. Apply Lemma 18 with $m = n^2$ and $f_z \coloneqq e^{\sigma L \phi_z}$. The left side of Lemma 18 is $\mathbb{E}_{\Omega_A} e^{\sigma \tilde{p}}$.

Since $\phi_z \in \{0,1\}$, $\mathbb{E}_Z f_z(Z) = 1 + (e^{\sigma L} - 1)\mathbb{E}_Z \phi_z(Z)$, and $\ln(1+y) \le y$ for $y > -1$. So the right side of Lemma 18 is at most $\exp\bigl(\chi_L(\sigma) \int \mathbb{E}_Z \phi_z(Z)\, d\mu(z)\bigr)$. Where $\zeta(z) \neq 0$, $G\zeta(z) \sim \mathcal{N}(0, I)$; so, by Fubini's theorem, $\int \mathbb{E}_Z \phi_z(Z)\, d\mu(z) = \mathbb{E}_{\Omega_A} \tilde{p}$. Hence $\mathbb{E}_{\Omega_A} e^{\sigma \tilde{p}} \le e^{\chi_{L'_A}(\sigma)\, \mathbb{E}_{\Omega_A} \tilde{p}}$, as claimed. □

Lemma 20. *On $\mathcal{G}$, the frame constants are of order $1/n$, i.e., $L_W \le \vartheta_W / n$ and $L_A \le \vartheta_A / n$ for explicit reals $\vartheta_W, \vartheta_A > 0$ that depend only on n; the Numerical certificates give examples of their values [3].*

Proof sketch for L_A. The full proof is in Lean. Fix $\mathcal{D} \in \mathcal{G}$, with $\tau_0 > 0$. Let $y \coloneqq \hat{c} + \tau_0 \xi$, so that $\tilde{x} = y/|y|$, let $w \coloneqq \tau_0^2 / |y|^2$, let $J(z)$ be the Jacobian of h at z, and let $\Pi_z \coloneqq I - zz^\top$. The sketch has three parts.

The first part of the sketch bounds L_A by a cutoff on S. Since $\zeta\zeta^\top = \nu_A h^\perp h^{\perp\top} / S$ and $\zeta\zeta^\top \preceq I$, for every real $d_0 > 0$,

$$L_A \le \frac{\nu_A}{\gamma_A d_0} \bigl(\|\Pi_{b_A} \operatorname{Cov}_{\tilde{x}}(h)\, \Pi_{b_A}\|_{\mathrm{op}} + |\Pi_{b_A} \mathbb{E}_{\tilde{x}} h|^2 \bigr) + \mathbb{P}_{\tilde{x}}(S < \gamma_A d_0), \tag{5}$$

with Π_{b_A} as in Lemma 3.

The second part bounds the covariance by the Gaussian Poincaré inequality (Boucheron et al., 2013, Theorem 3.20), which states that $\operatorname{Var} f(\xi) \le \mathbb{E}|\nabla f(\xi)|^2$ for continuously differentiable f. Lean proves the inequality for Lipschitz f, since h is Lipschitz but not continuously differentiable. As $y/|y|$ is not Lipschitz near $y = 0$, Lean applies it with $y / \max\{|y|, \upsilon\}$ in place of $y/|y|$, for $\upsilon > 0$, and lets $\upsilon \to 0$, using $\mathbb{E}|y|^{-4} < \infty$. For each $v \in \mathbb{R}^{n^2}$, the chain rule gives $\nabla_\xi \langle h(\tilde{x}), v \rangle = (\tau_0/|y|)\, \Pi_{\tilde{x}} J(\tilde{x})^\top v$ almost everywhere. Hence

$$\nu_A \operatorname{Cov}_{\tilde{x}}(h) \preceq D, \qquad D \coloneqq \mathbb{E}_\xi B B^\top, \qquad B \coloneqq \sqrt{\nu_A w}\, J(\tilde{x}) \Pi_{\tilde{x}}. \tag{6}$$

The third part bounds D by pairs of proxy inputs. Let T map f in $L^2(\xi; \mathbb{R}^n)$ to $\mathbb{E}_\xi B f(\xi)$. Then $\|D\|_{\mathrm{op}} = \|TT^*\|_{\mathrm{op}} = \sup_{\|f\|_2 = 1} |Tf|^2$, and $|Tf|^2$ is the average of $f(\xi)^\top B^\top B' f(\xi')$ over two independent proxy inputs, with B' the value of B at the second input. The Lean proof bounds this average by the Schur test (Halmos and Sunder, 1978, Theorem 5.2), with explicit charges for exceptional pairs of inputs. With this bound on $\|D\|_{\mathrm{op}}$, (6) bounds the covariance term of (5). The Lean proof bounds the other two terms of (5) on $\mathcal{G}$, and then give $L_A \le \vartheta_A / n$, as claimed. On the range $8{,}000 \le n \le 11{,}882$, they use a variant of (5). □

[1] `BstLean:Part157Hell,Part157Repl,Part156Master`; Parts 134, 137–138, 140, 143, 157, 159, 204–205, 210, 214, 218, 227, 232–233, 236; Parts 156–157, 192–193, 196–197, 203, 243, 261, 293. In Lean, $\bar{\Delta}$ is `HbarW`.

[2] `BstLean:Part157Gp,Part157Mgf,Part158Final,Part156Comp`. Lean proves Lemma 18 as `gaussGP`, by Ornstein–Uhlenbeck interpolation.

[3] `BstLean:Part159Hid,Part159Uncond,Part227Frame,Part236Frame,Part290Frame,Part293World` for L_W; `BstLean:Part160Split,Part188Poincare,Part151Poinc,Part182Schur,Part291WeightedFrame,Part293FrameAdapter` for L_A; Parts 134, 137–138, 140, 143, 157, 159, 204–205, 210, 214, 218, 227, 232–233, 236, Parts 105, 107, 109, 112, 134, 136, 148–149, 151–154, 157–158, 160, 188, 194–195, 198–200, 206, 208–209, 211, 215, 217, 219, 221–222, 229, 235, 238, 244–246, 273, Parts 290–293.

Assembly. Say that Est *errs* if it does not succeed. Following the chain (1) of the High-level overview, split the event that Est errs into three events: $\mathcal{D} \notin \mathcal{G}$; $\mathcal{D} \in \mathcal{G}$ and $|\sqrt{R} - \sqrt{\tilde{q}(\mathcal{D})}| > \Delta_s$; or neither occurs and Est still errs. By Lemma 17, the first event has probability at most $\delta_{\mathrm{hid}} + \mu_W / \Delta_W^2 + \delta_{\mathrm{ext}}$ over $\mathcal{D}$. By Lemma 16, the second event has probability at most $2e^{-K\Delta_s^2}$ over r. In the third event, let $Q \coloneqq \mathbb{E}_{\Omega_A} \tilde{p}$. Lemmas 14 and 16 give $|\sqrt{R} - \sqrt{Q}| \le \Delta_A + \Delta_s$, and Lemma 17 gives $|\sqrt{\mathbb{E}_{\Omega_W} p} - \sqrt{\tilde{p}}| \le \Delta_W$. Let $\hat{z}_+ \coloneqq \max(1/100, \Gamma \max(0, \sqrt{Q} - \Delta_A - \Delta_s)^2)$ and $\hat{z}_- \coloneqq \Gamma(\sqrt{Q} + \Delta_A + \Delta_s)^2/10$. By the definition of U, as in the proof of Lemma 15, Est then errs only if $p > \hat{z}_+$, or if $p < \hat{z}_-$ and $\hat{z}_- > 1/1000$. These are tail events of the two links marked "conc." in (1). Fix reals $\sigma_+ > 0 > \sigma_-$. Let $E_\pm$ be explicit functions of $(\Delta_A + \Delta_s, \Delta_W, \vartheta_W, \vartheta_A, \sigma_\pm)$ and of Γ, with no further dependence on n, and let $\mathcal{B}_\pm \coloneqq e^{nE_\pm}$. Only the Lean proof states $E_\pm$. Assume that $E_+ \le 0$, $E_- \le 0$,

$\Delta_A + \Delta_s + \Delta_W < 1/\sqrt{100\Gamma}$ and $2(\Delta_A + \Delta_s) \le 1/\sqrt{100\Gamma}$. For each $\mathcal{D} \in \mathcal{G}$, over (Ω_W, Ω_A), Lemmas 19 and 20, with the bound of Lemma 17, give probability at most $\mathcal{B}_+$ that $p > \hat{z}_+$, and at most $\mathcal{B}_-$ that $p < \hat{z}_-$ with $\hat{z}_- > 1/1000$. Hence the probability that Est errs is at most

$$\delta(n) \coloneqq \delta_{\text{hid}} + \mu_W/\Delta_W^2 + \delta_{\text{ext}} + \mathcal{B}_+ + \mathcal{B}_- + 2e^{-K\Delta_s^2} \text{ [1]}.$$

Lemma 21 below will show that, for every integer $n \ge 8{,}000$, Est errs with probability at most 0.01: by $\delta(n) \le 0.01$ if $n \ge 11{,}883$, and by a finer form of $\delta(n)$ if $n \le 11{,}882$. Hence $\mathbb{P}\big(p \le U \le \max(10p, 1/100)\big) \ge 0.99$ for every such n and every $t \in \mathbb{R}$, which is the success bound of the Beating Sampling Theorem.

[1] Parts 156–157, 192–193, 196–197, 203, 243, 261, 293. In Lean, the bound is `network_master` in `Part196Master`.

Numerical certificates.

Lemma 21. *For every integer $n \ge 8{,}000$ and every $t \in \mathbb{R}$,* Est *errs with probability at most* 0.01*. If $n \ge 11{,}883$, Lean gives explicit constants, which depend on n, that satisfy the assumptions of the Assembly and give $\delta(n) \le 0.01$. If $n \le 11{,}882$, the probability is at most $\delta_{\text{low}} < 0.00874$, for δ_{low} defined below.*

Lean proves Lemma 21 on four ranges of n: $n \ge 16{,}200$ [1]; $12{,}225 \le n \le 16{,}199$ [2]; $11{,}883 \le n \le 12{,}224$ [3]; and $8{,}000 \le n \le 11{,}882$ [4]. Lean checks the numerical certificates by exact rational enclosures, including Taylor bounds for exp. It splits the ranges into subranges, some of a single n, with their own constants. When a tail certificate covers several values of n with fixed exponent bounds, Lean checks it at the least of them and transfers it to larger n, since $E_\pm \le 0$ makes $\mathcal{B}_\pm$ nonincreasing in n. On the first three ranges, $\Delta_s = 10^{-5}$, so $2e^{-K\Delta_s^2} = 2e^{-100}$.

For example, for $16{,}384 \le n \le 16{,}500$, Lean uses the constants

$$\begin{aligned} &\Delta_W = 0.0024160420, \quad \Delta_A = 0.006583 + 10^{-20}, \quad \Delta_s = 10^{-5}, \\ &\vartheta_W = 1.377, \qquad\qquad \vartheta_A = 2.036, \qquad\qquad \sigma_+ = 0.1528, \quad \sigma_- = -0.4038, \end{aligned}$$

and checks

$$\begin{aligned} &E_+ \le -0.000364553074, \quad E_- \le -0.0003551354576, \\ &\mathcal{B}_+ + \mathcal{B}_- \le 0.005519, \qquad \delta_{\text{hid}} + \mu_W/\Delta_W^2 + \delta_{\text{ext}} \le 0.00448, \end{aligned}$$

so that $\delta(n) \le 0.00448 + 0.005519 + 2e^{-100} < 0.01$ [5].

On the last range, Lean uses a finer failure bound. Let $\mathcal{G}_{\text{low}}$ be the intersection of the six events on $\mathcal{D}$ that the proof for this range uses; it differs from $\mathcal{G}$. On $\mathcal{G}_{\text{low}}$, Lean uses the single values $\vartheta_A = 1.56$, $\Delta_A = 0.0064$ and $\Delta_s = 5 \cdot 10^{-6}$. In place of the one threshold Δ_W on $\bar{\Delta}$ and the one constant ϑ_W, it uses two nonnegative scalars ϖ_1, ϖ_2, which only the Lean proof defines, with $\varpi \coloneqq \varpi_1 + \varpi_2 \le 0.324425$ and $L_W \le (0.419 + \varpi)/n$. Lean covers the values of $\bar{\Delta}$ and ϖ by finite intervals. On each pair of intervals, it checks at $n = 8{,}000$ a bound on $\min(1, \mathcal{B}_+ + \mathcal{B}_-)$, with finer exponents in place of $E_\pm$; the same bound holds for every $n \ge 8{,}000$. Beyond the last interval for $\bar{\Delta}$, it uses the bound 1. It averages these bounds over $\mathcal{G}_{\text{low}}$ through moments of $\bar{\Delta}^2$, ϖ_1^2 and ϖ_2. The probability that Est errs is then at most

$$\delta_{\text{low}} \coloneqq 0.0026872 + 0.0057 + 0.0003431447924852 + 10^{-9} = 0.0087303457924852.$$

The first term bounds $\mathbb{P}(\mathcal{D} \notin \mathcal{G}_{\text{low}})$. The second and third terms together bound the averaged interval bounds, including the bound 1 beyond the last interval. The last term bounds $2e^{-K\Delta_s^2} = 2e^{-25}$ [6]. On $\mathcal{G}_{\text{low}}$, the frame bounds of Lemma 20 hold with $\vartheta_W = 0.743425$ and $\vartheta_A = 1.56$.

For odd $n \ge 8{,}001$, the values $\bar{h}_{g,i}$ and $\bar{h}^{(2)}_{g,i}$ of the finite rule of step 5 are within $10^{-90} n^{-10} a$ of $\mathbb{E}_V X_{g,i}$ and within $3 \cdot 10^{-90} n^{-10} a^2$ of $\mathbb{E}_V X_{g,i}^2$, for every real a with $|s_g \alpha_i| \le a$ and $0 \le \sqrt{1 - s_g^2}\, \alpha_i^\perp \le a$. For every odd $n \ge 8{,}001$, on the good event of the proof for the

range of n, Lean turns these errors into a change of at most 10^{-20} in $\sqrt{\tilde{q}}$, the fourth term in the proof of Lemma 14; for the example above, it is the term 10^{-20} in Δ_A [7].

[1] Parts 172–188, 190, 278.
[2] Parts 243, 249–260, 262–264, 266–271, 278, 285–287.
[3] Parts 275–277, 280–281.
[4] Parts 290–293.
[5] `BstLean:Part186GLedC,Part186GCombC,Part160Ledger2Gen,Part156Num`.
[6] `BstLean:Part293Ledger,Part293Assembly,Part293EndpointInterface,Part236Master`.
[7] `BstLean:Part278Err,Part278StBase3,Part281Odd,Part293Arithmetic`.

Total cost. The step costs sum to $2m_W^2+1024n^5+4m_W J_W+O(n^{4.5})+2Km_F^2+O(Knm_F)$, where $1024n^5$ is the cost of step 5 and $4m_W J_W$ that of the rule B_{J_W} of step 2; for $8{,}000 \le n \le 16{,}384$ the first is 22 to 27 times the second. The leading term $2m_W^2$ is the cost of the product $W_2\psi$ of step 2; one exact sample computes the product $W_2\,\mathrm{ReLU}(W_1 u)$, which has the same leading cost. The K evaluations of F cost about $2Km_F^2 \approx 10^{24}$ at $n = 8{,}000$, far below $2n^7 > 4 \cdot 10^{27}$. For every $n \ge 8{,}000$, the total is below $1.0003 \cdot 2n^7$ [1].

[1] `BstLean:Part282Total,Part282JoinCost,Part279ScheduleOut,Part289Cost`; Parts 150, 189, 279, 282, 288–289.

The cost model in Lean charges nothing for rational constants or for reading the seed. The Cost items of steps 1, 2, 3, 5 and 6 estimated this uncharged work. The sum of these estimates is below $2 \cdot 10^{18}$ at $n = 8{,}000$, less than 10^{-9} times $2n^7$. Almost all of the sum comes from the Gaussian draws of step 6. These estimates are not checked in Lean.

6 LLM usage

LLMs from multiple providers was used to discover, prove, and explain this result, via the following methodology.

Discovery via informal proof. LLMs produced the proofs under human direction. Claude Opus 5.5 (medium effort) coordinated the work, wrote and challenged paper proofs, and checked each result before the result was recorded. GPT-6.1-Sol (Codex, medium effort) challenged paper proofs. GPT-6-Astra (Codex, medium effort) wrote the proof for $8{,}000 \le n \le 11{,}882$. By volume of written reports the paper work was split roughly 57% Codex and 42% Opus. Each claim was challenged by a second model before the claim was relied on.

Formal proof. LLMs formalized the proofs in Lean. GPT-6.1-Sol formalized most of the Lean development, including the shared library and the certificates for $n \ge 12{,}225$. GPT-6-Astra (high effort) formalized the proof for $8{,}000 \le n \le 11{,}882$, the band-wise root certificates and the final assembly. Claude Opus 5.5 formalized the treatment of odd dimensions, the operation count, the sharp root bounds and several assemblies. Claude Sonnet 5.5 (medium effort) formalized some numerical tables. Of the 202 Lean parts in the import closure of the two final theorems, roughly 77% are Sol's, 16% Opus's, 6% Astra's and 1% Sonnet's. The Lean kernel checks the proof of the final statement. A Lean theorem can compile even when its statement is not the intended claim (cf. Ilin and Nugent, 2026, Sections 1 and 4); for that reason GPT-6-Astra (high effort) and Claude Opus 5.5 separately compared this note's text with the Lean statements of the setting, of Est and of the conclusion.

Explanation. The human author iteratively refined a rule-set for Claude Opus 5.5 improving the exposition, and made occasional direct changes, until the logic flow of the argument became sufficiently organized and clear to the author.

7 Related work

ARC's program. The Alignment Research Center (ARC) proposes to estimate the probability that a catastrophe detector fires on a model's outputs, by analyzing the model, because sampling is slow when failures are rare (Neyman et al., 2025; Winer, 2026): $\mathcal{K}$ samples are unlikely to show a failure much rarer than $1/\mathcal{K}$, and so cannot estimate its rate to within a constant factor. The network A, the detector C and the distribution of x play the roles of ARC's model, detector and input distribution, and the failure rate p of the Beating Sampling Theorem is the quantity ARC's scheme needs. ARC's *Matching Sampling Principle* is Conjecture 1 below. Let $M_\theta(c, x)$ be a model with weights θ, a random context c and a random input x; in this paragraph and the next, c and x are ARC's.

Conjecture 1 (Matching Sampling Principle, ARC)**.** *For every architecture M there is an estimator with the following property. For every θ there is an explanation π of size $O(|\theta|)$, obtained in some unspecified way, such that for every tolerance $\varepsilon > 0$ the estimator uses θ, π, c and ε to estimate the mean output $\mathbb{E}_x[M_\theta(c, x)]$, and: averaged over the context c, its mean squared error is at most ε^2 times the average variance $\mathbb{E}_c \operatorname{Var}_x M_\theta(c, x)$; it runs in time $O(\varepsilon^{-2})$ forward passes of M_θ; and it is mechanistic, a requirement that ARC leaves informal, i.e., it estimates deductively from the structure of M_θ instead of from sampled behavior.*

Averaging ε^{-2} independent samples of $M_\theta(c, x)$ achieves exactly this accuracy at exactly this cost, but is not mechanistic; this is why the principle speaks of *matching* sampling. Random weights can be placed in the context; ARC's results for random networks take this form and need no explanation, as the paragraph "Mechanistic estimation for random networks" will discuss.

Now let $M_\theta(c, x)$ be a failure indicator, equal to 1 when the detector fires and 0 otherwise, so that its mean is the failure rate p. A 0/1 variable with mean p has variance $p(1 - p)$, which is about p when failures are rare. As a benchmark, consider a single context with failure rate p and an error of about $\varepsilon\sqrt{p}$. This error is smaller than p itself only when $\varepsilon^2 \lesssim p$, so that the estimate is right to within a constant factor, and then the time allowed is about $1/p$ forward passes. That is also what sampling costs to estimate a probability p to within a constant factor. So Conjecture 1, as stated, does not promise that a mechanistic estimator beats sampling on rare failures; ARC hopes to do better on structured models. The program Est of the Beating Sampling Theorem beats naive-99% samplers on the target of Est, $p \le U \le \max(10p, 1/100)$: for every $p \le 1/1000$, the cost of Est is more than 447 times smaller than the expected cost of any naive-99% sampler (Section 4). We prove no such lower bound for samplers that need to succeed only for most networks, nor for importance sampling or splitting.

Mechanistic estimation for random networks. Wu et al. (2026) estimate the expected output of a wide random MLP under Gaussian input by propagating cumulants, without running the network. Their theorems assume fixed depth and polynomial activations as the width $n \to \infty$. For tolerances ε at least inverse-polynomial in n, they give mean squared error $O(\varepsilon^2)$ averaged over the weights. When $\varepsilon^2 = O(1/n^2)$, their running time is $O(n/\varepsilon^2)$, against $\Theta(n^2/\varepsilon^2)$ for Monte Carlo. Their experiments use ReLU at width 256 and beat sampling in error per FLOP, including on rare thresholded events. Whether such methods can match sampling at large depth is open there, and ARC's White-Box Estimation Challenge (Hilton et al., 2026; AIcrowd, 2026) asks contestants to estimate the same mean output, at width 256 and depth 32 in Phase 1 and width 1,024 and depth 16 in Phase 2, without requiring mechanistic methods. For one layer, Hilton (2025) deductively estimates the probability that all output neurons are active (an orthant probability), matching the mean squared error of sampling averaged over the weights, up to logarithmic factors in the running time. Compared with this work, the Beating Sampling Theorem covers only depth two and prescribed width scalings, but it concerns a probability over a non-Gaussian input distribution produced by a larger network, holds with explicit constants at every $n \ge 8{,}000$, and is checked in Lean.

Other approaches. Rare-event methods estimate failure probabilities of networks with fewer samples, by importance sampling or multilevel splitting (Webb et al., 2019; Wu and Hilton, 2025), but still evaluate the network many times and give no guarantee of the kind proved here. Moment propagation through ReLU layers under a Gaussian approximation is standard in Bayesian deep learning (Wu et al., 2019), with empirical checks but no error bounds. Sound probabilistic verifiers bound output probabilities rigorously by branch and bound over the input space (Boetius et al., 2025), for a given network, trained ones included; the experiments of Boetius et al. (2025) use input dimensions far smaller than $2n \geq 16{,}000$, and that paper gives no cost guarantee comparable to the Beating Sampling Theorem. Infinite-width theory (Matthews et al., 2018; Lee et al., 2018; Hanin, 2023) describes random networks over the random weights, whereas Est estimates a probability over inputs for one draw of the weights. Gross et al. (2024) prove compact accuracy bounds for small trained one-layer attention-only transformers, with proofs shorter than brute-force enumeration; our result instead bounds the cost of an estimator, for random networks at frontier input widths.

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A Network sizes

Table 2 repeats Table 1, with fuller notes in its caption. The input dimension of A is comparable to the embedding widths of the models in Table 2, but its hidden width and weight count are far larger: A has about n^4 weights, $4.1 \cdot 10^{15}$ at $n = 8{,}000$, while all of Kimi K3 has a reported $2.8 \cdot 10^{12}$ parameters (Moonshot AI, 2026). The failure metric F has hidden width $\lceil n^{1.5} \rceil = 715{,}542$ at $n = 8{,}000$, which is closer in size to one mixture-of-experts layer, i.e., to the values $524{,}288 \le h \le 2{,}752{,}512$ of Table 2. The transformer MLPs also differ in kind: they are trained, use gated activations rather than ReLU, have one hidden layer, and sit between attention layers, while A is a random dense ReLU network with two hidden layers.

Network	n	h	ℓ	n'	Parameters
Kimi K3 MLPs (routed experts) (Moonshot AI, 2026)	3,584	2,752,512	1	3,584	$2.8 \cdot 10^{12}$
GLM-5.3 MLPs (Z.ai, 2026)	6,144	524,288	1	6,144	$7.53 \cdot 10^{11}$
DeepSeek V4 Pro MLPs (DeepSeek-AI, 2026)	7,168	1,179,648	1	7,168	$1.6 \cdot 10^{12}$
Qwen3.8-2.4T-A95B MLPs (Qwen Team, 2026)	8,192	1,048,576	1	8,192	$2.4 \cdot 10^{12}$
Muse Spark Open MLPs (Meta, 2026)	4,096	1,048,576	1	4,096	$8.16 \cdot 10^{11}$
Beating Sampling Theorem, Agent (A)	**8,000**	**64,000,000**	**2**	**8,000**	$\mathbf{4.1 \cdot 10^{15}}$
Beating Sampling Theorem, World (W)	**16,000**	$\mathbf{4.58 \cdot 10^{13}}$	**2**	**8,000**	$\mathbf{2.1 \cdot 10^{27}}$

Table 2: The agent and world networks of the Beating Sampling Theorem at $n = 8{,}000$, and the MLP blocks of large open-weight transformers, read from their published configurations (October 2026). The last column counts the parameters of the whole model: the weights of A and of W, and the totals stated in each model card or release announcement; GLM-5.3's model card states no total, so its entry is the parameter count that Hugging Face reports for its checkpoint. Kimi K3's routed experts act on a 3,584-dimensional projection of its 7,168-dimensional embedding, and Muse Spark Open's on a 4,096-dimensional projection of its 8,192-dimensional embedding. The mixture-of-experts layers have 896 routed experts of width 3,072 with 16 used per token (Kimi K3), 512 of width 2,048 with 10 per token (Qwen3.8), 384 of width 3,072 with 6 per token (DeepSeek V4 Pro), 256 of width 4,096 with 8 per token (Muse Spark Open) and 256 of width 2,048 with 8 per token (GLM-5.3). All but Muse Spark Open also have one or two shared experts. The configuration of Muse Spark Open sets no shared experts. The official Muse Spark Open checkpoint is gated, so its figures are from the configuration of a public copy of the checkpoint and from the Transformers implementation.

B Lean files

The extracted project has 5,833 modules: the import closure of the modules of the two Lean theorems that state the bounds of the Beating Sampling Theorem, `Part293.classifier_stated2_8000` and `Part289.progOut_run2`, together with two modules, `Part288Audit` and `Part289All`, that contain only imports and axiom-audit commands. Of these modules, 4,290 define a declaration that the proofs of these two theorems use. The table assigns every module to one group (groups S0–S10 follow Section 5: S0 is the setting, S1 the definition of Est, S2 and S3 what is known after step 4, S4 what is known after step 6, S5 the assembly, S6–S9 the numerical certificates on the four ranges of n, and S10 the total cost; S11 holds modules that are compiled because they are imported, but contribute no declaration to these proofs). A script checks that the groups cover every module. Parts are module-name prefixes `BstLean.Part<k>`.

Step	Group	Parts	Files	Used	Role
S11	G00	29, 33, 35–46, 48–53, 56–64, 66, 69, 71–74, 78, 80–82, 84, 86, 88–90, 93, 105–109, 112, 130–135, 138–145, 149–154, 156–180, 182, 185–188, 192–194, 197, 199–201, 204–205, 208–213, 216–217, 220–223, 235–236, 238–239, 241–245, 247, 249, 251–257, 260–262, 264–266, 269, 271, 273, 275–276, 278–279, 281–282, 287–290, 292–293	1543	0	These imported modules contribute no declaration to either final theorem.
S0	G01	130, 132, 156–157	11	11	The definitions specify the Gaussian networks, normalized input, target probability, and success event.
S0	G02	29, 33, 35–36, 41, 48, 53, 56, 58, 73, 80, 105–108, 131, 135, 139, 144, 146	41	41	Gaussian and spherical integration supplies moments, symmetry, measurability, and polar coordinates.

Step	Group	Parts	Files	Used	Role
S0	G03	130, 132–133, 142, 145	24	24	The ReLU kernel and layer moment bounds control the means and covariances of hidden vectors.
S0	G04	130, 140, CostModel	3	3	The program language charges one operation for each arithmetic operation, square root, and ReLU evaluation, and counts the operations of matrix-vector products.
S1	G05	157, 169, 272	14	14	The program computes the two centres and the height tables and draws the Gaussian mixtures; the closed forms of the height moments are proved for even dimensions.
S1	G06	141, 150, 159, 201	31	31	Finite binomial sums and truncated series approximate the Gaussian ReLU means and the spherical constants.
S1	G07	278	11	11	The arithmetic rule for odd dimensions approximates the exact height moments and bounds the resulting table perturbation.
S1	G08	289	2	2	The scale floor equals 0.17 below dimension 11,883 and 0.20 at or above it before multiplication by $2/n$.
S2	G09	134, 157	5	5	Gaussian regression separates each last weight matrix into its retained action on a centre and an independent residual.
S2	G10	137, 140, 143, 204, 210	23	23	The computed World centre and residual scale approximate the corresponding input averages.
S2	G11	138, 157, 159	18	18	Gaussian Hellinger bounds transfer the World centre and scale errors to square roots of event probabilities.
S2	G12	205, 214, 218	68	68	Conditional kernel expansions bound the World mean, covariance, direction error, and hidden energy fluctuations.
S2	G13	227, 233	22	22	World event bounds control the centre, noise window, frame, and integrated Hellinger error.
S2	G14	232, 236	14	14	Integration over radius and frame intervals converts radius and frame moments into a bound on the classifier failure probability.
S3	G15	107, 109, 148, 151–152, 244, 246	35	35	Gaussian concentration, Poincaré inequalities, and matrix tail bounds control norms and quadratic forms.
S3	G16	105, 112, 134, 136	14	14	Gaussian correlation estimates and second-moment identities bound the operator norm of averaged residual directions.
S3	G17	157–158	14	14	The Gaussian Brascamp–Lieb inequality converts a frame bound into an exponential-moment bound for an event probability.
S3	G18	151, 153, 229	42	42	Jacobian and inverse-radius estimates give covariance and moment bounds for the Agent hidden vector.
S3	G19	149, 154, 160, 194, 198, 206, 211	86	86	Mean and deviation estimates bound the error and energy of the computed Agent centre.
S3	G20	160, 195, 215, 222	47	47	Covariance, centre error, and residual-energy tails combine into the Agent frame bound.

Step	Group	Parts	Files	Used	Role
S3	G21	188	9	9	A capped reciprocal weight and the Poincaré inequality control the average of squared residual directions.
S3	G22	199, 208, 235, 238	40	40	Schur bounds split input pairs into central, intermediate, and exceptional regions to control the Agent covariance.
S3	G23	200, 209, 217, 221, 245, 273	50	50	Conditional Gaussian pair estimates control the tangent Jacobian and its error after the first Agent layer.
S3	G24	219	22	22	Rotation and moment bounds control the input gradient of the hidden vector's coordinate along the computed Agent centre.
S4	G25	161–165	45	45	Hellinger and chi-square bounds compare location-scale mixtures and separate fluctuation from bias.
S4	G26	166–168, 212	56	56	Height-conditioned Jacobian moments bound the variation of the Agent location and residual scale.
S4	G27	169–171, 202, 213, 216, 220, 223, 239	130	130	Moment bounds for the finite height tables control interpolation error, location bias, and scale bias.
S4	G28	228, 241	38	38	Gaussian height tails and variable bias parameters bound the Hellinger error of Agent replacement.
S4	G29	207, 242, 248	35	35	Conditional variance and reciprocal expansions reduce the Hellinger error of the Agent mixture.
S4	G30	289	8	8	The bias and mixture estimates remain valid for the smaller positive scale floor.
S5	G31	156–157, 196, 203	20	20	The two residual exponential-moment bounds, the replacement root bound, and the sampling error imply the required two-sided output guarantee.
S5	G32	193, 197, 261	13	13	Exact interval arithmetic and finite series certify the scalar inequalities in the final probability bounds.
S5	G33	192, 243	5	5	Parameterized World, Agent, and endpoint bounds connect the analytic estimates to the network classifier.
S5	G34	293	2	2	The four dimension ranges combine into the success theorem for every integer $n \geq 8{,}000$.
S6	G35	172–181	142	142	Explicit bounds on exceptional probabilities and Hellinger errors establish the classifier guarantee at large dimensions without an upper dimension limit.
S6	G36	182–188, 190	386	386	Sharper covariance, energy, and table bounds extend the upper-range guarantee down to dimension 16,200.
S6	G37	278	45	45	The odd-dimension table perturbation transfers the upper-range guarantee to the stated arithmetic program.
S7	G38	250, 260, 268	312	312	World event and Hellinger certificates supply the individual dimensions in the lower portion of the middle range.
S7	G39	252, 257, 262, 266	352	352	Bounds on exceptional Agent events and height table errors control the Hellinger error of Agent replacement in the middle range.

Step	Group	Parts	Files	Used	Role
S7	G40a	251, 253–254, 256, 258–259, 263	160	160	Local noise intervals supply centre, direction, energy, and covariance inputs for the middle-range frame bounds.
S7	G40b	255, 264	327	327	Checked Gaussian matrix and geometric bounds supply the local covariance and frame estimates below dimension 12,350.
S7	G40c	269	432	432	Local covariance, centre, and energy estimates bound the frame for dimensions 12,350 through 13,224.
S7	G41a	268	175	175	Finite noise-cell tables certify the scalar frame inequalities on dimensions 12,350 through 13,224.
S7	G41	243, 249, 268	76	76	Rational endpoint certificates verify the scalar success inequalities for dimensions 12,225 through 13,224.
S7	G42	267, 270, 278, 286	51	51	The row certificates give the stated-program guarantee on dimensions 12,225 through 13,199; group G45 assembles the rows through 13,224.
S7	G43	271	35	35	Uniform World, centre, and moment bounds supply the upper portion of the middle dimension range.
S7	G44	285	48	48	Interval estimates control the probability of a table error and the Hellinger error of Agent replacement in the upper middle range.
S7	G45	287	24	24	The lower row certificates and upper interval certificates give contiguous coverage from dimension 12,225 to 16,199.
S8	G46	275, 277	13	13	Parameterized frame and Hellinger bounds support the local estimates on dimensions 11,883 through 12,224.
S8	G47	276, 280	83	83	Sharper table moments and exceptional-probability bounds control the Hellinger error of Agent replacement on dimensions 11,883 through 12,224.
S8	G48	281	99	99	World certificates bound the hidden events and Hellinger error on dimensions 11,883 through 12,224.
S8	G49a	281	101	101	Rational cell and endpoint certificates verify the frame and failure bounds on dimensions 11,883 through 12,224.
S8	G49b	281	37	37	The World, Agent, and scalar bounds assemble the stated classifier guarantee on dimensions 11,883 through 12,224.
S8	G49	281	85	85	Local centre, energy, and covariance bounds supply the frame estimates on dimensions 11,883 through 12,224.
S9	G50	290, 293	17	17	Uniform World bounds control the noise window, exceptional probabilities, and Hellinger moments throughout the low band.
S9	G51	291, 293	36	36	Uniform Agent centre and height-conditioned moments supply the location and residual-scale estimates on the low band.
S9	G52	291, 293	99	99	Weighted Schur and Poincaré estimates with checked interval covers bound the Agent frame throughout the low band.

Step	Group	Parts	Files	Used	Role
S9	G53	292–293	67	67	A common event with bounded complement probability controls the stored location and energy errors throughout the low band.
S9	G54	293	30	30	Bias and conditional-mixture estimates at the positive scale floor bound the Hellinger error of Agent replacement throughout the low band.
S9	G55	293	13	13	A radius and frame cover at dimension 8,000 bounds the failure probability uniformly over the low band.
S9	G56	293	11	11	The World, Agent, table, and frame estimates combine with the endpoint bound and exceptional-probability bounds to prove success throughout dimensions 8,000 through 11,882.
S10	G57	189, 279	12	12	Explicit operation schedules bound the World pass, Agent tables, Gaussian-mixture draws, and final output.
S10	G58	150, 282, 288	21	21	Counted arithmetic programs return the stated centres, tables, samples, and output and include the comparisons in the table search.
S10	G59	289	3	3	The revised program returns the stated output and costs less than $1.0003 \cdot 2n^7$ operations for every $n \geq 8{,}000$.
	Total		5833	4290	